Intuition Engine Architecture

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Intuition Engine is a multi-CPU fantasy computer with 6 heterogeneous CPU cores, 6 video systems, audio engines and players, a copper coprocessor, DMA blitter, and extensive I/O peripherals - all connected through a unified MachineBus. Total guest RAM is sized at boot from platform-dispatched usable-RAM detection (`/proc/meminfo` on Linux, `GlobalMemoryStatusEx` on Windows, and `hw.memsize` on Darwin) minus a per-platform reserve. Darwin RAM sizing uses a page-aligned conservative half of `hw.memsize` as the detected base before applying the per-platform reserve. Each CPU/profile sees an active visible RAM clamped to its own ceiling. Guest software discovers sizes through the SYSINFO MMIO pairs (`SYSINFO_TOTAL_RAM_LO/HI`, `SYSINFO_ACTIVE_RAM_LO/HI`) and IE64 `CR_RAM_SIZE_BYTES`. This document describes the system architecture with diagrams showing chips, buses, internal functional units, and data flow paths.

The diagrams below describe wired runtime behaviour. Platform availability follows the runtime dispatch path; for example, the Z80 JIT is available on amd64 builds only.

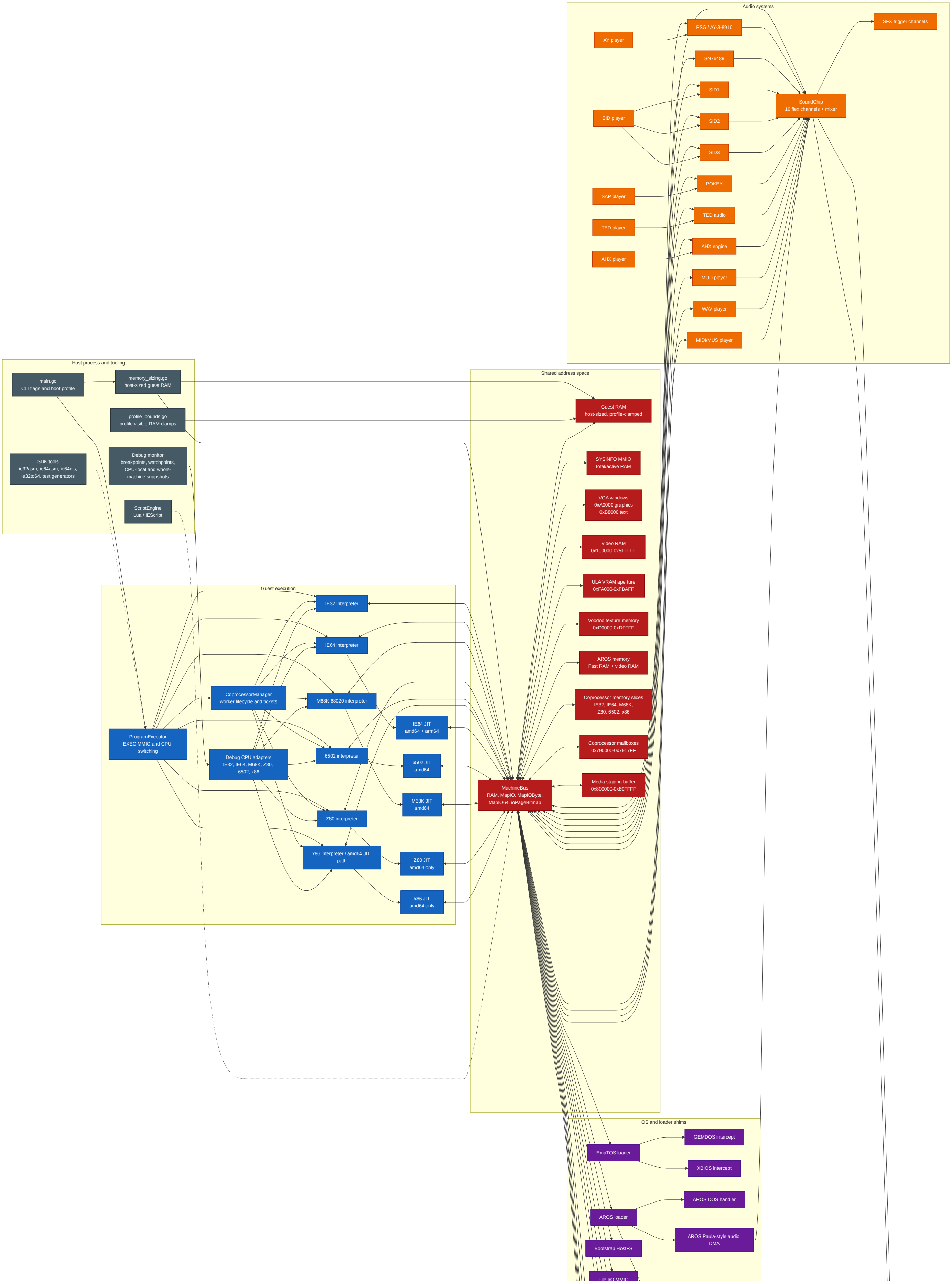
Reading the Architecture Tables and Diagrams

Boxes represent runtime components. Arrows represent operational paths: direct calls, bus mappings, shared-memory paths, queues, or host-service dependencies. Memory-map rows describe guest-visible address ranges and the device or subsystem that owns each range.

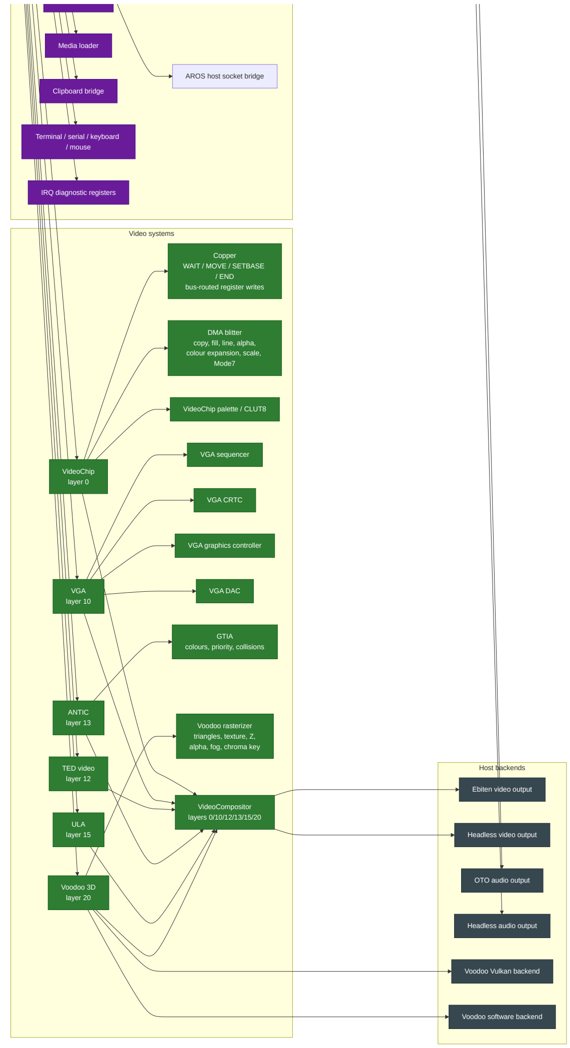
Diagram/table item	Meaning
Address range	Guest-visible range plus decoded mapping, owner, and reservation/use semantics
Visible RAM size	RAM visible to the current CPU/profile, as reported by SYSINFO and CPU-specific discovery paths
CPU or JIT entry	CPU core or JIT backend available through the runtime dispatch path
Audio, video, or peripheral box	Engine, player, device, or bridge that participates in the running machine
Debugger or scripting path	Monitor, debug adapter, or Lua binding path used for inspection or automation

If a feature is platform-limited, the table or diagram names the supported platform path instead of implying that every implementation file is active in every build.

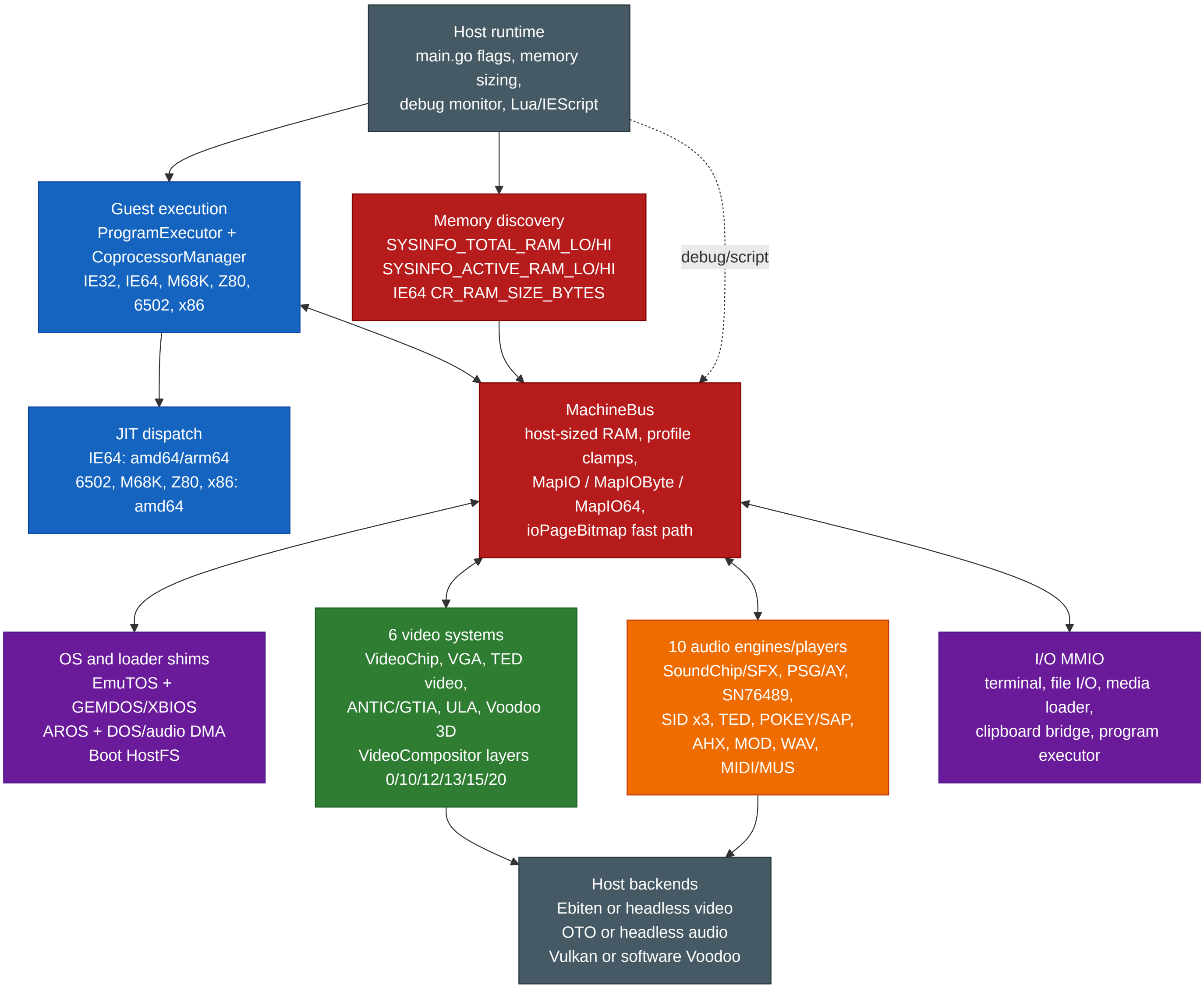
Single Complete Architecture Diagram



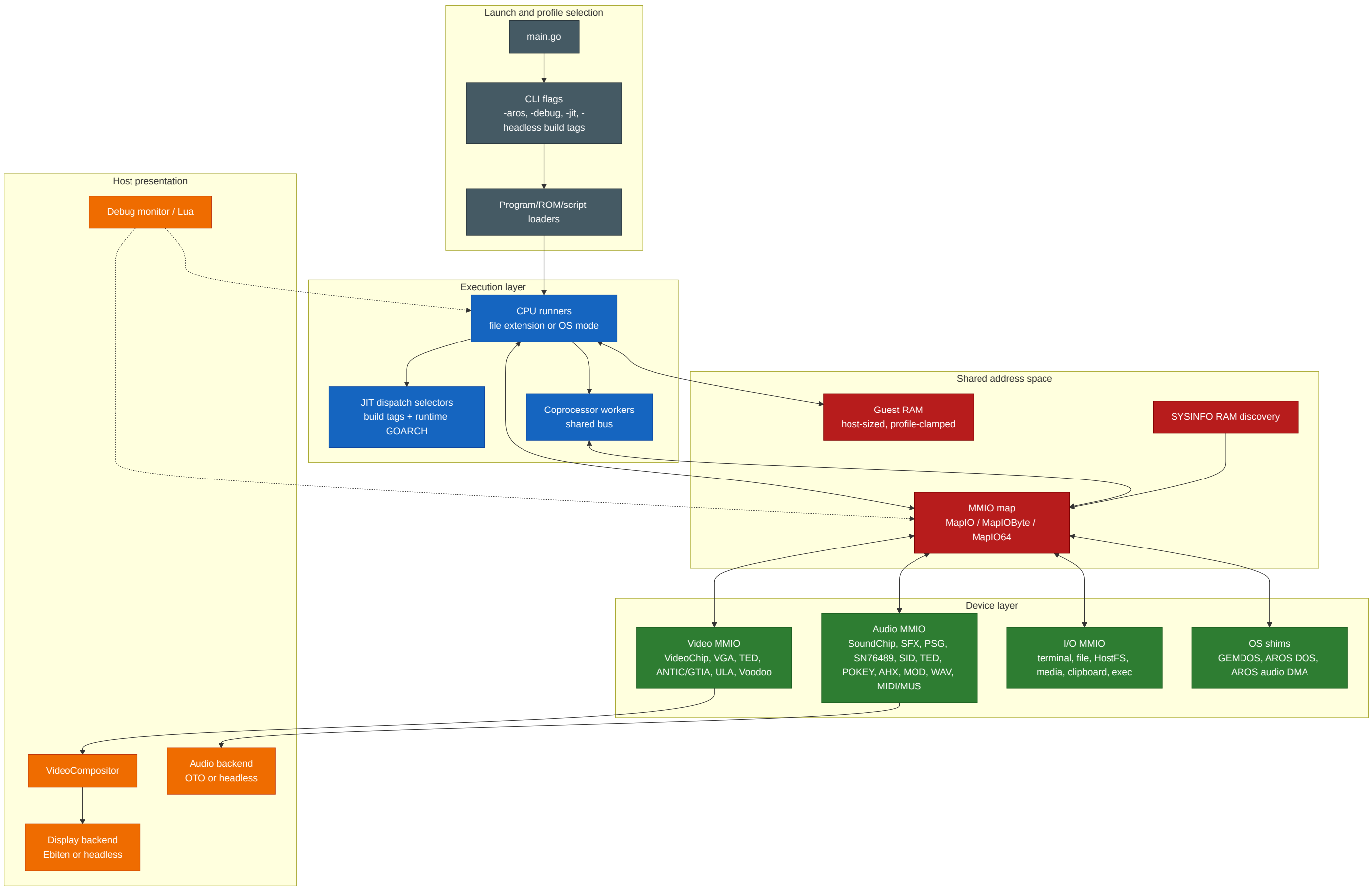
Audio systems
AY player
PSG / AY-3-8910
SFX trigger channels
SN76489
SID1
SID player
SoundChip
10 flex channels + mixer
SID2
SID3
POKEY
SAP player
TED audio
TED player
AHX engine
AHX player
MOD player
WAV player
MIDI/MUS player
Host process and tooling
main.go
CLI flags and boot profile
memory_sizing.go
host-sized guest RAM
profile_bounds.go
profile visible-RAM clamps
SDK tools
ie32asm, ie64asm, ie64dis, ie32to64, test generators
Debug monitor
breakpoints, watchpoints, CPU-local and whole-machine snapshots
ScriptEngine
Lua / IEScript
Shared address space
Guest RAM
host-sized, profile-clamped
SYSINFO MMIO
total/active RAM
VGA windows
0xA0000 graphics
0xB8000 text
Video RAM
0x100000-0x5FFFFF
ULA VRAM aperture
0xFA000-0xFBAFF
Voodoo texture memory
0xD0000-0xDFFFF
AROS memory
Fast RAM + video RAM
Coprocessor memory slices
IE32, IE64, M68K, Z80, 6502, x86
Coprocessor mailboxes
0x790000-0x7917FF
Media staging buffer
0x800000-0x80FFFF
MachineBus
RAM, MapIO, MapIOByte, MapIO64, ioPageBitmap
Guest execution
IE32 interpreter
IE64 interpreter
CoprocessorManager
worker lifecycle and tickets
M68K 68020 interpreter
IE64 JIT
amd64 + arm64
ProgramExecutor
EXEC MMIO and CPU switching
Debug CPU adapters
IE32, IE64, M68K, Z80, 6502, x86
6502 interpreter
6502 JIT
amd64
M68K JIT
amd64
Z80 interpreter
x86 interpreter / amd64 JIT path
Z80 JIT
amd64 only
x86 JIT
amd64 only
OS and loader shims
GEMDOS intercept
EmuTOS loader
XBIOS intercept
AROS DOS handler
AROS loader
AROS Paula-style audio DMA
Bootstrap HostFS
File I/O MMIO



1. Whole-System Architecture


Host runtime
main.go flags, memory sizing,
debug monitor, Lua/IEScript
Guest execution
ProgramExecutor + CoprocessorManager
IE32, IE64, M68K, Z80, 6502, x86
Memory discovery
SYSINFO_TOTAL_RAM_LO/HI
SYSINFO_ACTIVE_RAM_LO/HI
IE64 CR_RAM_SIZE_BYTES
debug/script
JIT dispatch
IE64: amd64/arm64
6502, M68K, Z80, x86: amd64
MachineBus
host-sized RAM, profile clamps,
MapIO / MapIOByte / MapIO64,
ioPageBitmap fast path
OS and loader shims
EmuTOS + GEMDOS/XBIOS
AROS + DOS/audio DMA
Boot HostFS
6 video systems
VideoChip, VGA, TED video,
ANTIC/GTIA, ULA, Voodoo 3D
VideoCompositor layers 0/10/12/13/15/20
10 audio engines/players
SoundChip/SFX, PSG/AY, SN76489,
SID x3, TED, POKEY/SAP,
AHX, MOD, WAV,
MIDI/MUS
I/O MMIO
terminal, file I/O, media loader,
clipboard bridge, program executor
Host backends
Ebiten or headless video
OTO or headless audio
Vulkan or software Voodoo

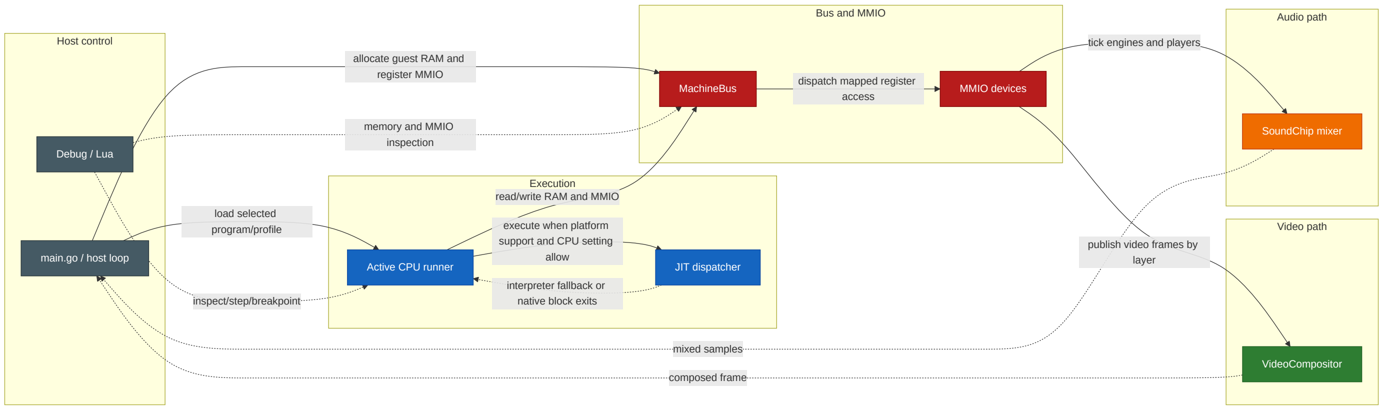
2. Layered System Overview



Bus architecture notes:

- **Concurrent multi-CPU bus -** the Program Executor selects the primary CPU mode, but the Coprocessor Manager can launch additional worker CPUs that run concurrently on the same bus. Address dispatch is immutable after mapping is sealed, `ioPageBitmap` provides the fast path, and I/O callbacks protect their own mutable state. There is no central bus-arbitration model exposed to guest software.
- **No centralised interrupt controller -** each CPU has per-CPU interrupt lines (IRQ/NMI as `atomic.Bool`). Peripherals signal the active CPU directly.
- **MMIO dispatch -** the bus uses an `ioPageBitmap []bool` fast path (page = 256 bytes). Non-I/O pages use direct unsafe pointer access with zero dispatch overhead.

Runtime Data and Control Flow



Subsystem Matrix

Subsystem	Runtime surface	Primary files	Wired registration / dispatch
CPU cores	IE32, IE64, M68K, Z80, 6502, x86	`cpu_*.go`, `cpu_*_runner.go`	`main.go` selects runners by file extension, OS mode, or EXEC MMIO
JIT	IE64 on amd64/arm64; 6502, M68K, Z80, x86 on amd64	`jit_dispatch.go`, `jit_6502_dispatch.go`, `jit_m68k_dispatch.go`, `jit_z80_dispatch.go`, `jit_x86_dispatch.go`	Build tags plus `runtime.GOARCH`; non-supported hosts use dispatch stubs
Bus and RAM	Host-sized guest RAM, profile clamps, MMIO, byte/64-bit handlers	`machine_bus.go`, `memory_sizing.go`, `profile_bounds.go`, `sysinfo_mmio.go`	`main.go` registers devices before execution; `MachineBus.SealMappings` prevents late maps
Machine lifecycle	Load resolution, reset quiesce, CPU/profile recreation, monitor/runtime rewiring	`machine_lifecycle.go`, `main.go`	`main.go` owns concrete devices; `Machine` applies reset/load orchestration through injected dependencies and profile targets
Video	VideoChip, VGA, TED video, ANTIC/GTIA, ULA, Voodoo	`video_chip.go`, `video_vga.go`, `video_ted.go`, `video_antic.go`, `video_ula.go`, `video_voodoo.go`	`main.go` maps each register/VRAM block and registers compositor layers 0/10/12/13/15/20
Audio	SoundChip/SFX, PSG/AY, SN76489, SID x3, TED, POKEY/SAP, AHX, MOD, WAV, MIDI/MUS	`audio_chip.go`, `sfx_trigger.go`, `psg_engine.go`, `sn76489_chip.go`, `sid_engine.go`, `ted_engine.go`, `pokey_engine.go`, `ahx_player.go`, `mod_player.go`, `wav_player.go`, `midi_player.go`	`main.go` maps chip/player MMIO and registers sample tickers/mixers into SoundChip; register-mapped players share `PlayerControlState` for staged playback requests
OS integration	EmuTOS, AROS, GEMDOS/XBIOS, AROS DOS, Paula-style DMA	`emutos_loader.go`, `aros_loader.go`, `gemdos_intercept.go`, `aros_dos_intercept.go`, `aros_audio_dma.go`	OS modes install intercept MMIO and loader state during boot/reset
Tooling	Assemblers, disassembler, transpiler, generators	`assembler/`, `internal/ie64meta/`, `cmd/gen_ie64_opmeta/`, `cmd/ie32to64/`, `cmd/gen_m68k_cputest/`, `cmd/gen_interp6502/`	Makefile builds SDK tools into `sdk/bin/`; IE64 opcode constants and name tables are generated from metadata

Internal Ownership Notes

- **IE64 opcode metadata -** `internal/ie64meta/table.go` is the source of truth for generated IE64 opcode constants and mnemonic/name tables used by the runtime CPU, monitor disassembler, standalone assembler, standalone disassembler, and internal assembler package. Build commands for `ie64asm` and `ie64dis` build `./assembler` with the relevant tag so generated files are compiled with the tagged tool entry point.
- **Machine lifecycle -** `Machine` in `machine_lifecycle.go` owns reset/load sequencing through dependency fields and profile targets. It preserves the existing guest-visible reset contract while keeping quiesce, failed-load rollback, CPU recreation, profile loading, and monitor/runtime rewiring out of the host event loop.
- **Audio playback control -** register-mapped music players use `PlayerControlState` for staged pointer/length registers, optional high pointer, bus reads, busy/error/loop state, subsong selection, and async generation invalidation. This covers SID, PSG/SN/SNDH, TED, POKEY/SAP, AHX, MOD, WAV, and MIDI/MUS without changing their public MMIO contracts.
- **Video scheduling -** `VideoScheduler` centralises ticker ownership for the compositor and migrated video render-loop shims. Tests can drive it manually; migrated engine files do not own `time.NewTicker` directly.
- **Voodoo software wrappers -** headless and `novulkan` builds share `softwareVoodooBackend` forwarding. The tagged Voodoo files keep feature registration and constructor selection only, so their `VoodooBackend` method sets stay identical.
- **IE64 MMU walk sharing -** CPU translation and bootstrap hostfs guest-pointer translation share the page-table walk result (`MMUSharedWalkResult`). CPU policy applies trap causes and A/D updates; hostfs policy requires `PTE_U` and leaves A/D bits untouched.

Platform JIT Matrix

The host-side JIT support is intentionally asymmetric and follows the dispatch files, not emitter-file presence. Release amd64 builds target x86-64-v3 (`GOAMD64=v3`, the Makefile default) for the best Go-compiler codegen (AVX2, BMI1/2, FMA), but this is a recommendation, not a hard requirement: the JIT only *requires* SSE4.1, which it emits unconditionally for IE64 FINT (ROUNDSS). The amd64 backend detects this at runtime via CPUID; on a host without SSE4.1 `initJIT` (via `checkJITHostFeatures`) declines to enable the JIT and the CPU falls back to the interpreter rather than aborting, so plain `go run .` / `go build` at the default `GOAMD64=v1` works on any x86-64 CPU (older hosts simply run interpreted). AVX2/BMI2/LZCNT paths are separately runtime-gated via `x86Host` and are never emitted unless the host supports them.

Host platform	JIT-enabled guest cores	Dispatch files
Linux amd64	IE64, 6502, M68K, Z80, x86	`jit_dispatch.go`, amd64 per-core dispatch files
Linux arm64	IE64	`jit_dispatch.go`; Z80 dispatch compiles but keeps `z80JitAvailable` false
Windows amd64	IE64, 6502, M68K, Z80, x86	amd64 per-core dispatch files
Windows arm64	IE64	IE64 dispatch only; per-core non-IE64 stubs
macOS amd64	IE64, 6502, M68K, Z80, x86	amd64 per-core dispatch files
macOS arm64	IE64	IE64 dispatch plus Darwin arm64 JIT write-protect helpers

On macOS amd64, the JIT reuses the shared x86-64 host backends. On macOS arm64, executable memory uses the native `MAP_JIT` model with thread-pinned write protection toggles, and non-IE64 guest cores remain interpreter-only on arm64 hosts.

IE64 JIT 64-bit Execution Model

The IE64 JIT (amd64 and arm64) executes code, accesses data, and operates the stack at any `uint64` virtual or physical address, matching the interpreter. There is no `uint32` truncation on the PC, data addresses, stack pointer, branch targets, or chain targets. The one stack exception is the amd64 non-MMU fused-leaf path (documented below): it raw-indexes `[MemBase+SP]` and so assumes the SP is in the low window.

`SEI64` and `CLI64` are emitted as per-instruction interpreter bails (not NOPs) so they mutate `interruptEnabled` under the JIT. External device interrupts are delivered through a record-only sink and a pending mask polled at instruction and block boundaries by both the interpreter and the JIT dispatcher; see the "External Interrupt Delivery" section of this manual for the full model.

- **Low memory window**: each guest sees a contiguous low RAM window backed by the dense `cpu.memory[]` slice, capped at 256 MiB for the IE64 family (`lowMemWindowBytes`). The actual `len(bus.memory)` is `min(autodetectedTotalGuestRAM, busMemCap)` (main.go), so on a small host the window can be smaller than the cap. Addresses above the window resolve through the bus's 64-bit physical path, backed by the `Backing` interface -- production IE64 boots on Linux/darwin bind high RAM via `MmapBacking`; `SparseBacking` is the test implementation. High physical addresses are supported for code fetch and for data / stack *operands* (the latter via the helper exit, including a high SP), with one caveat below.
- **High-PC stack-op caveat**: a block *fetched from* high physical backing that itself contains a stack op (`PUSH`/`POP`/`JSR`/`RTS`/`JSR_IND`) is not JIT-compiled -- `ExecuteJIT` takes the `highPhys && containsStackOp(instrs)` path and runs the block one instruction at a time through `interpretOne()` (which routes the stack through the bus). This is pinned by `TestJIT_HighPC_StackOpBlock_BailsToInterpreter`. Stack ops in blocks fetched from the low window are JIT-compiled and, for a high SP, route through the helper exit -- **except** the fused-leaf path below.
- **Fused-leaf high-SP exception**: on amd64 with the MMU off, `scanBlock` may fuse a `JSR64` to a small leaf subroutine (`ie64FusedJSRLeafCall` / `ie64FusedRTSLeafReturn`, gated by `ie64ScanJSRLeafFusionEnabled` -- amd64 only). The fused emit (`jit_emit_amd64.go`) handles the call/return push/pop as **raw** `[MemBase+SP]` traffic *before* the normal `emitJSR_AMD64`/`emitRTS_AMD64` high-SP guards, so it does **not** take the helper exit. This is safe only because the marker is suppressed whenever `mmuBail` is set, but `mmuBail` for fused leaves is set by `compileBlockMMU` (MMU mode) only -- in non-MMU mode a fused leaf with an SP outside the low window would read/write past `cpu.memory[]`. Non-MMU guests are therefore expected to keep the stack in the low window; high-SP correctness via the helper exit holds for unfused stack ops, not for fused leaves.
- **Direct fast path vs. helper exit**: native code uses a direct `[memBase+addr]` access only when the MMU is off **and** the address is in the low window (size-aware bound: `addr <= MemSize - accessBytes`). Otherwise it takes the JITContext-mediated *helper exit*.
- **Helper exit protocol**: native code cannot safely call back into Go (it runs on g0 via `asmcgocall`), so on a high/MMU access it writes a structured request to the `JITContext` (`NeedHelper`, `HelperAddr`, `HelperSize`, `HelperVal`, `HelperRd`, `HelperPC`, `LiveSP`), flushes the live SP and the faulting instruction's PC, and returns through the epilogue. `ExecuteJIT` dispatches the request (`handleJITHelper`) and re-enters the JIT. Helper ops cover `LOAD`/`STORE`, `FLOAD`/`FSTORE`, `DLOAD`/`DSTORE`, `PUSH`/`POP`, and the control-flow `JSR`/`RTS`/`JSR_IND`.
- **Interpreter parity**: helpers use `cpu.loadMem`/`storeMem` for data and `cpu.mmuStackWrite`/`mmuStackRead` for the stack -- the same paths as the interpreter. JSR sets PC to the call target, RTS sets PC to the popped return address, and JSR_IND sets PC to `rs + sext(imm32)` -- the displacement is sign-extended, so a negative `imm32` (bit 31 set) subtracts (with `rs == R31` based on the decremented SP). The FP load/store side effects differ: FLOAD and DLOAD set the destination FP register **and** the FP condition codes; FSTORE and DSTORE read the FP register and write memory, leaving FP registers and condition codes unchanged -- matching the interpreter in each case.
- **MMU rule**: when the MMU is enabled, non-atomic data/stack ops take the helper exit (virtual-to-physical mapping is not a direct offset). The dispatcher refreshes `ctx.MMUEnabled` before every `callNative` so the native guards are

never stale.

- **Fault contract**: the handler sets `cpu.PC = HelperPC` before the operation, so `trapFault` records the correct `faultPC`; faulting instructions are not counted and re-execute after the trap. Pre-decrement ops (PUSH/JSR/JSR_IND) roll the SP back on a trapping fault.
- **Atomics carve-out**: atomics (CAS/XCHG/FAA/FAND/FOR/FXOR) do not use the helper-exit protocol. The native emitters run sequentially-consistent host-atomic fast paths only for aligned, non-MMU, low-window RAM addresses; MMU-on, high-address, MMIO, or unaligned cases bail to the interpreter so the canonical `atomicRMW64` trap and bus semantics apply. `compileBlockMMU` also sets `mmuBail` on fused JSR/RTS leaf markers, so under MMU the `OP_JSR64`/ `OP_RTS64` emit takes the `mmuBail` branch to `emitBailToInterpreter` (a whole-instruction interpreter bail with its own fault/accounting path) -- neither the raw fused fast path nor the guarded helper exit runs. In non-MMU mode `mmuBail` is unset, so the raw fused fast path is used (see the fused-leaf high-SP exception above).

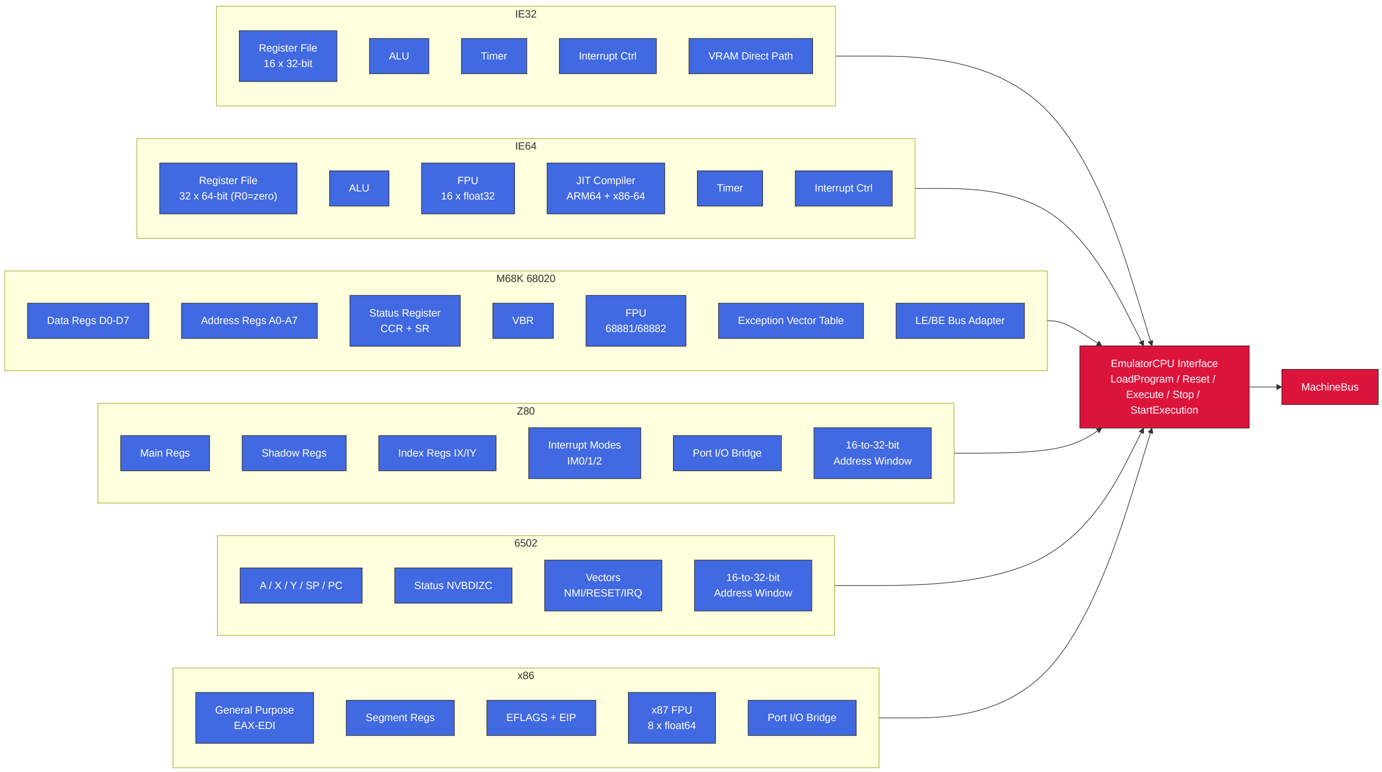
Build Profiles and Observable Runtime

Build tags change host backends, not the guest-visible ISA contract. The main guest bus, CPU cores, MMIO register addresses, assemblers, and script binding names remain the reference surface unless a specific backend is absent from the build. On amd64, release profiles target x86-64-v3 for codegen quality; lower `GOAMD64` levels still build and run, with SSE4.1 as the enforced runtime floor.

Profile	Build tags / knobs	Runtime effect visible to users
Default VM	no special tag	Ebiten display, Oto audio, Vulkan-backed Voodoo where available, and normal host integrations.
`novulkan`	`-tags novulkan`	Voodoo uses the software backend and does not require the Vulkan SDK. Guest Voodoo registers remain mapped.
`headless`	`-tags headless`	Display, audio backend, overlay, clipboard, and GUI integrations use stubs suitable for CI. CPU, bus, MMIO, scripting, and most device state paths still compile for tests.
`headless-novulkan`	`CGO_ENABLED=0 -tags "novulkan headless"`	Pure-Go portable VM build with headless stubs and software Voodoo path.

Headless stubs should be treated as backend substitutes, not as a different machine model. A test can still write video or audio MMIO and inspect guest state, but there is no host window, host audio device, or GUI overlay to observe the result directly.

3. CPU Subsystem



CPU Timers (IE32 / IE64)

Both IE32 and IE64 use CPU-internal countdown timers backed by atomic fields on the CPU structs.

- IE32 increments an internal cycle counter once per decoded instruction step, after fetch and operand resolution and before the decoded instruction body executes; when that counter reaches `SAMPLE_RATE`, it resets the counter and decrements the timer count by one.
- IE64 decrements the timer count once per decoded instruction step, after fetch/decode and before the decoded instruction body executes. When the IE64 timer is armed, JIT dispatch uses the CPU single-instruction path so an interrupt can be delivered before the body of the instruction whose decoded step expires the timer.
- On expiry, it sets timer state to expired and can trigger a CPU interrupt.
- If still enabled after interrupt handling, the count auto-reloads from the configured period.

There is no stable bus/MMIO timer control ABI at present. Legacy include symbols such as `TIMER_CTRL/TIMER_COUNT/TIMER_RELOAD` are retained for compatibility but are deprecated.

CPU Selection by File Extension

Extension	CPU	Address Space	Notes
`.iex` / `.ie32`	IE32	32-bit flat (clamped to active visible RAM)	Native RISC, 8-byte fixed instructions
`.ie64`	IE64	64-bit (sees full active visible RAM)	64-bit RISC, R0=zero, JIT on ARM64 + x86-64
`.ie68`	M68K	32-bit flat (clamped to active visible RAM)	Bare 68020, big-endian with LE bus adapter
`.ie65`	6502	16-bit + bank windows	Bank windows reach the banked-CPU visible ceiling
`.ie80`	Z80	16-bit + bank windows + port I/O bridge	Bank windows reach the banked-CPU visible ceiling

Extension	CPU	Address Space	Notes
`.ie86`	x86	32-bit flat (clamped to active visible RAM) + compatibility bank windows	Ordinary addressing is flat; bank/VRAM windows are compatibility overlays
`.tos` / `.img`	M68K	EmuTOS profile (`EmuTOS_PROFILE_TOP`)	EmuTOS boot with GEMDOS intercept
`.ies`	Script	N/A	Lua scripting engine (IE Script)

Bare `.ie68` uses the active-visible RAM ceiling; EmuTOS and AROS M68K loader modes use profile bounds.

AROS boot is selected by CLI mode (`-aros`, optional `-aros-image`), not file extension.

Z80 Port I/O Bridge

Z80 `IN`/`OUT` instructions are translated to bus MMIO accesses by the `Z80BusAdapter`:

Z80 Port	Chip	Bus Target	Protocol
`$A0-$AD`	VGA	`0xF1000`	Direct register map (MODE, STATUS, CTRL, SEQ, CRTC, GC, DAC, DAC read index, DAC mask, VRAM bank)
`$B0-$B7`	Voodoo 3D	`0xF8000`	Addr lo/hi + 4 data bytes (write to `$B5` triggers 32-bit bus write)
`$D0/$D1`	POKEY	`0xF0D00`	Register select / data
`$D4/$D5`	ANTIC	`0xF2100`	Register select / data (x4 stride)
`$D6/$D7`	GTIA	`0xF2140`	Register select / data (x4 stride, collision regs through `0xF21F8`)
`$E0/$E1`	SID	`0xF0E00/0xF0E30/0xF0E50`	Register select / data; select bits 5-6 choose SID1/SID2/SID3
`$E4/$E5`	SN76489	`0xF0C30/0xF0C31`	Data write / last-written read and ready-status read
`$F0/$F1`	PSG	`0xF0C00`	Register select / data
`$F2/$F3`	TED	`0xF0F00` / `0xF0F20-0xF0F6B`	Register select / data (audio indices `$00-$05`, video indices `$20-$32` x4 stride)
`$FE/$FD/$BE/$FA/$FB/$FC`	ULA	`0xF2000-0xF2014`	Border, control, status, VRAM address latch low/high, and paged VRAM data

Z80 16-bit Memory Translation

Z80 memory accesses in `0xF000-0xFFFF` are translated to bus `0xF0000-0xF0FFF`. ANTIC/GTIA access is provided through the Z80 port bridge (`$D4-$D7`) rather than general memory-mapped addressing.

x86 Port I/O Bridge

x86 shares the Z80 Voodoo port mapping (`$B0-$B7`) and most sound-chip port mappings, but **POKEY uses direct port-to-register mapping** (not the Z80's select/data protocol). x86 does not implement standard PC VGA I/O ports; VGA access is through the shared bus MMIO aperture and the direct `$A0000-$AFFFF` VRAM memory window.

x86 Port	Chip	Bus Target	Protocol
`$B0-$B7`	Voodoo 3D	`0xF8000`	Addr lo/hi + 4 data bytes

x86 Port	Chip	Bus Target	Protocol
`$60-$69`	POKEY	`0xF0D00+(port-0x60)`	**Direct** - port offset maps to writable register
`$D4/$D5`	ANTIC	`0xF2100`	Register select / data (x4 stride)
`$D6/$D7`	GTIA	`0xF2140`	Register select / data (x4 stride, collision regs through `0xF21F8`)
`$E0/$E1`	SID	`0xF0E00/0xF0E30/0xF0E50`	Register select / data; select bits 5-6 choose SID1/SID2/SID3
`$F0/$F1`	PSG	`0xF0C00`	Register select / data
`$F2/$F3`	TED	`0xF0F00` / `0xF0F20-0xF0F6B`	Register select / data (audio indices `$00-$05`, video indices `$20-$32` x4 stride)
`$FE`	ULA	`0xF2000`	Border colour only (bits 0-2)

Key difference from Z80: x86 POKEY access is direct - ports `$60-$69` map one-to-one onto writable POKEY registers at `0xF0D00+(port-0x60)`. ANTIC (`$D4/$D5`) and GTIA (`$D6/$D7`) keep their own select/data pairs.

x86 also directly accesses VGA VRAM at `$A0000-$AFFFF` in the memory path (no port translation needed).

Bank Windows (Z80 / 6502 / x86)

The banked 6502/Z80 ABI uses bank windows to access a 32 MiB banked-CPU visible ceiling from a 16-bit address space; bank translation rejects addresses at or above that ceiling. x86 ordinary addressing remains flat 32-bit, but the x86 bus adapter also implements the same compatibility bank windows plus a VRAM bank window. Those x86 window translations are capped at the legacy `DEFAULT_MEMORY_SIZE` ceiling; ordinary x86 addressing remains clamped by the active visible RAM exposed through the shared bus.

CPU Address	Size	Purpose	Bank Select Register
`$2000-$3FFF`	8KB	Bank 1 (sprite data)	`$F700/$F701` (lo/hi)
`$4000-$5FFF`	8KB	Bank 2 (font data)	`$F702/$F703` (lo/hi)
`$6000-$7FFF`	8KB	Bank 3 (general)	`$F704/$F705` (lo/hi)
`$8000-$BFFF`	16KB	VRAM window	`$F7F0` (bank number)
`$F000-$FFF9`	4090B	I/O window, excluding 6502 vectors	Hardwired: `$Fxxx` -> bus `$F0xxx`
`$F200-$F24F`	80B	Coprocessor gateway subrange	Hardwired: -> bus `$F2340+offset`
`$FFFA-$FFFF`	6B	6502 vectors (NMI/RESET/IRQ)	Identity mapped

6502 I/O Chip Page Dispatch

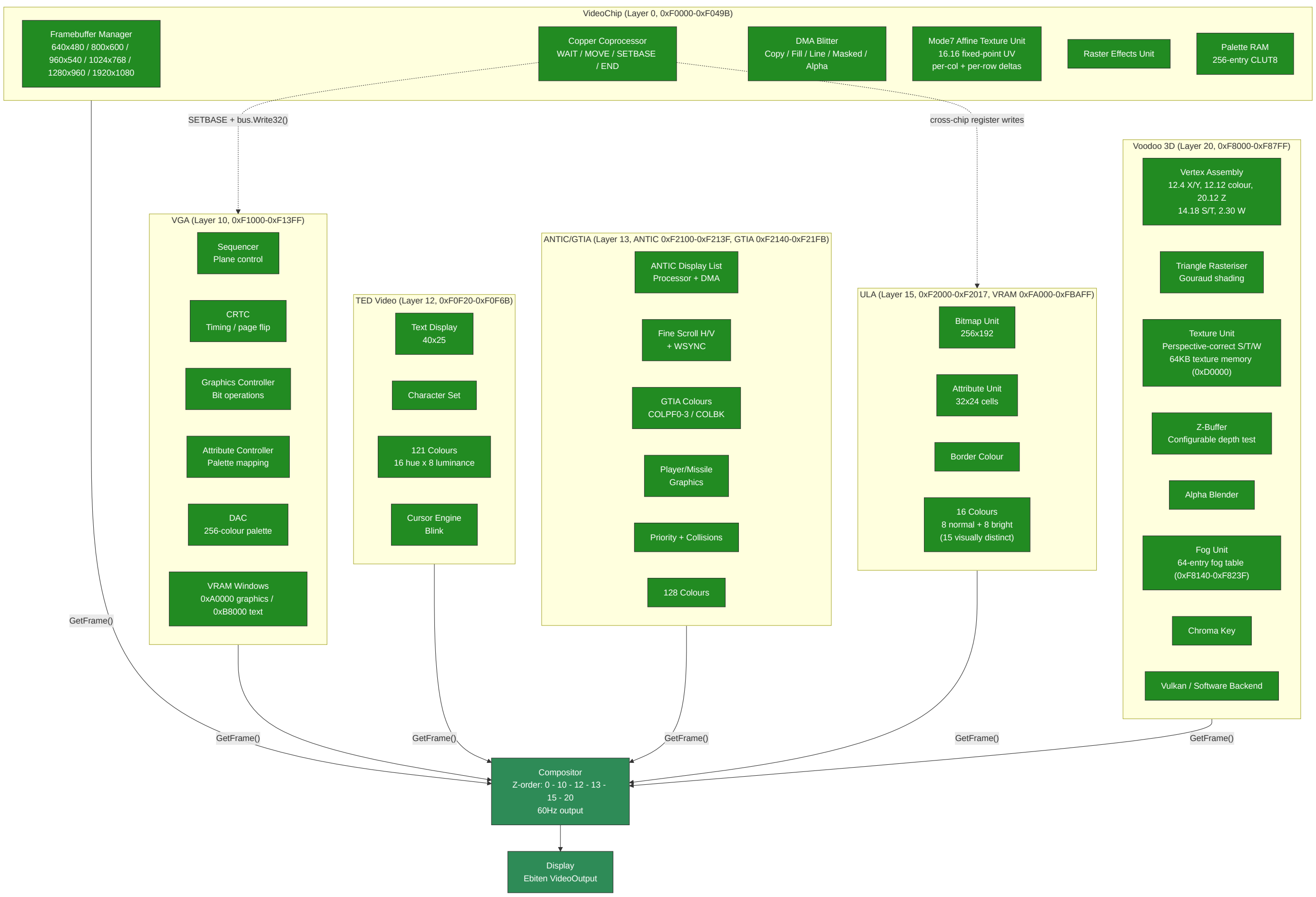
The 6502 uses `ioTable[page]` to route memory-mapped I/O through the bus:

6502 Address	Chip	Bus Target	Notes
`$D200-$D209`	POKEY	`0xF0D00+offset`	
`$D400-$D40F`	PSG	`0xF0C00+offset`	
`$D500-$D51F`	SID1	`0xF0E00+offset`	32-byte window
`$D520-$D53F`	SID2	`0xF0E30+offset`	32-byte window
`$D540-$D55F`	SID3	`0xF0E50+offset`	32-byte window

6502 Address	Chip	Bus Target	Notes
`$D600-$D605`	TED Audio	`0xF0F00+offset`	
`$D620-$D632`	TED Video	`0xF0F20+offset x4`	Stride-4 register mapping including raster compare registers
`$D700-$D70D`	VGA	`0xF1000`	Direct handler call plus DAC read index, DAC mask, and VRAM bank
`$D800-$D817`	ULA	`0xF2000+offset`	Registers plus paged VRAM data port

ANTIC/GTIA intentionally has no 6502 `$D400/$D000` compatibility surface; `$D400-$D40F` is PSG on the 6502 map.

4. Video Subsystem



Copper Cross-Chip Bus Access

The copper coprocessor is internal to VideoChip but can write to any MMIO-mapped chip on the bus:

- **Default**: MOVE targets VideoChip's own registers (`copperIOBase = VIDEO_REG_BASE`)
- **SETBASE opcode** changes `copperIOBase` to any MMIO address (e.g. `VGA_BASE 0xF1000`)
- Subsequent MOVE instructions compute `regAddr = copperIOBase + (regIndex * 4)` and route via `bus.Write32()` to VGA, ULA, or any other chip's MMIO
- This enables per-scanline palette changes, mode switches, and register manipulation on any video chip
- `copperIOBase` resets to `VIDEO_REG_BASE` at the start of each frame
- The compositor's `ScanlineAware` interface orchestrates this: `StartFrame()` -> `ProcessScanline(y)` -> `FinishFrame()`

Extended Blitter: BPP Modes, Draw Modes, Colour Expansion, and Scale

The blitter supports two pixel formats via `BLT_FLAGS` (`0xF0488`): RGBA32 (4 bpp, default) and CLUT8 (1 bpp). Bits 4-7 select one of 16 raster draw modes (Clear, And, Copy, Xor, Invert, etc.) applied per pixel during FILL and COPY operations. `BLT_OP=4` performs source-over alpha blending with source alpha in bits 31-24 using `out = (src*a + dst*(255-a))/255`. With `BLT_FLAGS` bit 12 (`ALPHA_TMPL`) set, `BLT_OP=4` instead treats `BLT_SRC` as an 8 bpp alpha plane (row stride in `BLT_SRC_STRIDE`) and blends the `BLT_FG` colour over the destination with the same source-over equation. This serves antialiased glyph and icon rendering (the AROS driver uses it for `PutAlphaTemplate`). `BLT_OP=7` performs nearest-neighbour scaling in RGBA32 or CLUT8. When `BLT_FLAGS=0`, the blitter defaults to Copy mode with RGBA32 for full backward compatibility.

Masked copies (`BLT_OP=3`) copy source pixels only where a 1-bit mask is set. The mask base address is in `BLT_MASK`, the mask row stride in `BLT_MASK_MOD` (0 selects packed rows of `(width+7)/8` bytes), and `BLT_MASK_SRCX` gives the starting bit offset within each mask row. Mask bits are sampled LSB-first by default; setting `BLT_FLAGS` bit 11 (`MASK_MSB`) selects MSB-first sampling, which matches the Amiga PLANEPTR convention used by AROS layer masks, so the driver can pass `CopyBoxMasked` masks to the hardware without conversion.

`BLT_CTRL` bit 0 starts the synchronous blit, bit 1 is read-only busy, and bit 2 enables a completion pulse on `IntMaskBlitter`. `BLT_STATUS` bit 0 is ERR, bit 1 is DONE, and bit 2 is sticky IRQ_PENDING (write 1 to clear). Invalid opcodes, out-of-range Mode7 samples, overflowed blitter bounds, and destination rectangles outside CPU-visible writable memory set ERR and do not silently fall back to COPY or wrap into another address range.

The colour expansion operation (`BLT_OP=6`) renders 1-bit glyph templates into coloured pixels for hardware-accelerated text. It reads a template from `BLT_MASK`, uses `BLT_FG`/`BLT_BG` (`0xF048C`/`0xF0490`) as foreground/background colours, and supports three modes: JAM2 (opaque - set bits write FG, clear bits write BG), JAM1 (transparent - only set bits write FG), and Invert (set bits XOR the destination). `BLT_MASK_MOD` (`0xF0494`) sets the template row stride and `BLT_MASK_SRCX` (`0xF0498`) provides sub-byte bit alignment for glyph fragments. Template bits are MSB-first (Amiga convention).

Line drawing (`BLT_OP=2`) supports an extended mode when `BLT_FLAGS != 0`: `BLT_DST` becomes the framebuffer base address, `BLT_WIDTH` holds the packed endpoint coordinates `(y1<<16)|x1`, and `BLT_DST_STRIDE` sets the row stride. This allows line drawing into arbitrary bitmaps (not just the active framebuffer) with BPP awareness and all 16 draw modes. When `BLT_FLAGS=0`, legacy behaviour is preserved (endpoint in `BLT_DST`, base at `VRAM_START`). In extended mode the blitter does not clip - callers must provide pre-clipped coordinates (the AROS driver uses Cohen-Sutherland clipping before calling the blitter).

Scale blits (`BLT_OP=7`) use `BLT_SRC`/`BLT_DST` as base addresses, `BLT_WIDTH`/`BLT_HEIGHT` as the source size in pixels, and `BLT_COLOR` as a packed destination size (`height << 16 | width`). `BLT_SRC_STRIDE` defaults to source width times bytes-per-pixel. `BLT_DST_STRIDE` defaults to the active VideoChip mode stride for VRAM destinations or destination width times bytes-per-pixel for ordinary memory. In non-direct VRAM mode, visible VRAM accesses use the active VideoChip front buffer and off-screen VRAM aperture accesses fall back to bus memory so back buffers can be presented through `VIDEO_FB_BASE`. Out-of-bounds source or destination rectangles, rectangles crossing the VRAM aperture end, zero source dimensions, and zero destination dimensions set `BLT_STATUS.ERR`.

Raster-band register draws (`VIDEO_RASTER_Y`, `VIDEO_RASTER_HEIGHT`, `VIDEO_RASTER_COLOR`, `VIDEO_RASTER_CTRL`) target the active framebuffer source. In RGBA32 mode, a nonzero `VIDEO_FB_BASE` writes the band into bus memory at that base; otherwise it writes the internal front buffer or direct VRAM slice. In CLUT8 mode, raster bands write indexed bytes at `VIDEO_FB_BASE`.

Register	Address	Description
`BLT_FLAGS`	`0xF0488`	BPP (bits 0-1), draw mode (bits 4-7), JAM1/invert flags (bits 8-10), MSB-first mask sampling (bit 11), alpha-template blend (bit 12)

Register	Address	Description
BLT_FG	0xF048C	Foreground colour for colour expansion
BLT_BG	0xF0490	Background colour for colour expansion
BLT_MASK_MOD	0xF0494	Template row modulo (bytes per row)
BLT_MASK_SRCX	0xF0498	Starting X bit offset in template

Mode7 Affine Texture Unit

The blitter's Mode7 operation (`bltOpMode7`) implements SNES-style affine texture mapping as a DMA blitter mode within VideoChip. It uses 8 dedicated MMIO registers (`0xF0058-0xF0074`):

Register	Address	Description
BLT_MODE7_U0	0xF0058	Starting U coordinate (16.16 fixed-point)
BLT_MODE7_V0	0xF005C	Starting V coordinate (16.16 fixed-point)
BLT_MODE7_DU_COL	0xF0060	U increment per column (16.16)
BLT_MODE7_DV_COL	0xF0064	V increment per column (16.16)
BLT_MODE7_DU_ROW	0xF0068	U increment per row (16.16)
BLT_MODE7_DV_ROW	0xF006C	V increment per row (16.16)
BLT_MODE7_TEX_W	0xF0070	Texture width mask (must be power-of-2 minus 1)
BLT_MODE7_TEX_H	0xF0074	Texture height mask (must be power-of-2 minus 1)

The rasteriser walks each destination pixel, computes the source UV from the affine matrix (origin + column delta + row delta), wraps via power-of-2 bitmask, and samples the source texture. This enables rotation, scaling, and perspective-like effects on tiled backgrounds - the same technique used by the SNES PPU2 for its Mode 7 background layer.

VideoChip Mode7 honours the `BLT_FLAGS` BPP field: RGBA32 samples and writes 4-byte pixels, while CLUT8 samples and writes 1-byte palette indices with BPP-aware default strides. In RGBA32 mode the default source stride is `(texture mask width + 1) * 4`; in CLUT8 mode the default source stride is `(texture mask width + 1)`. The destination stride follows the same BPP-aware default-stride rules as the other blitter operations: active framebuffer width in bytes for VRAM destinations, or destination width times bytes-per-pixel for ordinary memory.

Video Compositor

The compositor collects immutable frame snapshots from all enabled video sources and blends them in Z-order (layer 0 at the back, layer 20 at the front). For IEVideoChip CLUT8 mode, both mapped VRAM and direct bus-backed VRAM are converted through the palette before compositing. Desktop startup uses a 1920x1080 fullscreen presentation by default, and the x64 live-image launcher locks it fullscreen through `IE_LIVE_IMAGE=1`. Native sources keep their requested dimensions. The default native mode is 960x540 for exact 2x 1080p presentation. Video compositor default scale mode is stretch-fill; F11 toggles non-16:9 sources to aspect-fit. `Shift+F11` toggles fullscreen/windowed mode when that mode is not locked.

Two rendering paths:

1. **Scanline-aware path -** used when at least one enabled source implements `ScanlineAware`. The compositor advances scanline-capable sources in sorted layer order for each scanline, then blends all enabled sources in the

global layer order. Opaque full-frame sources can sit below, between, or above scanline-aware sources without breaking copper/VGA per-scanline effects.

2. **Full-frame fallback** - used when no enabled source is scanline-aware. It collects complete frames and blends them in sorted layer order. Same-size frame blending uses parallel goroutines with 60-line strips via `sync.WaitGroup`.

All-zero frame pixels are transparent; any nonzero alpha or RGB value is opaque. During compositing, zero-alpha nonzero-RGB pixels are promoted to opaque `0xFFRRGGBB` before they overwrite the destination. The compositor tick remains fixed at 60 Hz for guest VBlank compatibility; `GetRefreshRate()` reports the output backend rate, while `GetTickRate()` reports the compositor tick.

Triple-Buffer Protocol

All video systems except VideoChip use a lock-free triple-buffer protocol for `GetFrame()`:

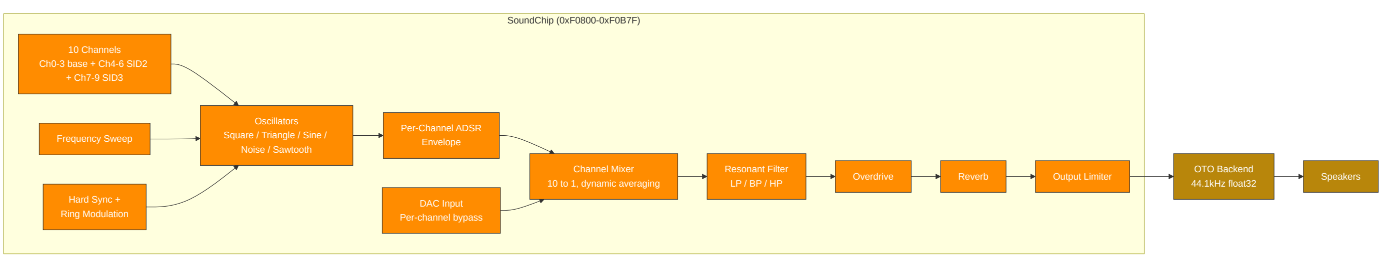
```
Slots:  writeIdx = 0 (producer-owned)
        sharedIdx = 1 (atomic, in-transit)
        readingIdx = 2 (consumer-owned)

Producer (render goroutine, after rendering into frameBufs[writeIdx]):
    writeIdx = sharedIdx.Swap(writeIdx)

Consumer (compositor, GetFrame):
    readIdx = sharedIdx.Swap(readIdx)
    return frameBufs[readIdx]
```

Important: `GetFrame()` performs an atomic Swap - calling it twice in a row swaps back to the previous buffer. Always call once and save the result.

On resolution change, all 3 buffer slots are reallocated and indices reset to `writeIdx=0`, `sharedIdx=1`, `readingIdx=2`.

5. Audio Subsystem

Synthesis Pipeline



SoundChip Dual-Interface Architecture

The SoundChip exposes two register interfaces for its channels:

Legacy per-waveform interface (`0xF0900-0xF09FF`) - dedicated register blocks with two decoded sweep-register exceptions:

Decoded ranges: `0xF0900-0xF093F except 0xF0914 and 0xF0918`, `0xF0940-0xF097F plus 0xF0914`, and `0xF0980-0xF09BF plus 0xF0918`.

Range	Channel	Default Waveform
`0xF0900-0xF093F`	Ch 0	Square, except `0xF0914` is triangle sweep and `0xF0918` is sine sweep
`0xF0940-0xF097F` plus `0xF0914`	Ch 1	Triangle
`0xF0980-0xF09BF` plus `0xF0918`	Ch 2	Sine
`0xF09C0-0xF09FF`	Ch 3	Noise
`0xF0A00-0xF0A6F`	-	Sawtooth + modulation/effects

Within the modulation/effects window, `0xF0A00-0xF0A1C` holds the sync-source and ring-modulation-source registers, `0xF0A20-0xF0A6F` holds the sawtooth legacy registers, and `0xF0A40`, `0xF0A50`, and `0xF0A54` hold overdrive and reverb controls.

FLEX unified interface (`0xF0A80-0xF0B7F`) - 4 channels with identical 64-byte register blocks. Each channel can be any waveform type:

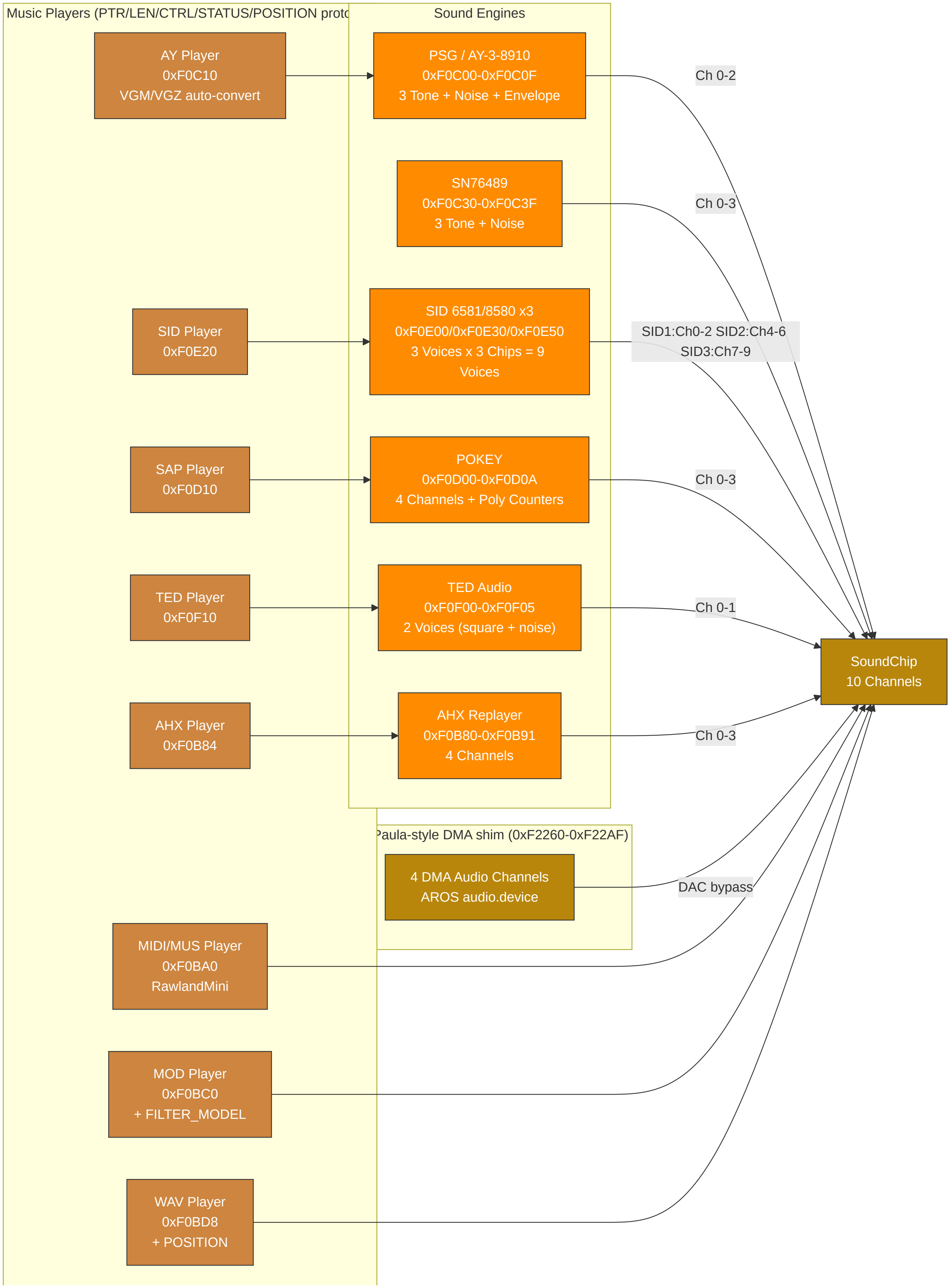
Offset	Register	Description
`+0x00`	`FREQ`	16.8 fixed-point Hz (value / 256.0)
`+0x04`	`VOL`	Volume (0-255)
`+0x08`	`CTRL`	Enable (bit 0), gate (bit 1)
`+0x0C`	`DUTY`	Pulse width duty cycle; high byte is PWM depth scaled by 1/256
`+0x10`	`SWEEP`	Frequency sweep rate
`+0x14-0x20`	`ATK/DEC/SUS/REL`	ADSR envelope
`+0x24`	`WAVE_TYPE`	Waveform selection
`+0x28`	`PWM_CTRL`	Pulse width modulation
`+0x2C`	`NOISEMODE`	Noise mode (0=white, 1=periodic, 2=metallic, 3=psg)
`+0x30`	`PHASE`	Phase reset
`+0x34`	`RINGMOD`	Ring modulation (bit 7=enable, 0-2=source)
`+0x38`	`SYNC`	Hard sync (bit 7=enable, 0-2=source)
`+0x3C`	`DAC`	DAC mode bypass (signed 8-bit sample, -128=-1.0, +127=+1.0)

FLEX channels are at `FLEX_CH0_BASE = 0xF0A80`, stride = `0x40`. Primary channels 0-3 occupy `0xF0A80-0xF0B7F`; SID2 flex channels 4-6 occupy `0xF0C40-0xF0CFF`; SID3 flex channels 7-9 occupy `0xF0D40-0xF0DFF`. `AUDIO_CTRL` uses bit 0 for enable and bit 1 for freeze. `ENV_SHAPE` remains at `0xF0804` for channel 0, with per-channel shapes at `0xF0860 + channel*4`. Both legacy and FLEX interfaces write to the same underlying 10-channel mixer. The FLEX interface is preferred for new code - the legacy interface exists for backward compatibility.

Filter and Modulation

The SoundChip's global resonant filter (`0xF0820-0xF0830`) supports low-pass, band-pass, and high-pass modes with cutoff frequency, resonance, and optional filter modulation source/amount registers. Cutoff and resonance smooth toward register targets at the same 0.02 coefficient used by per-channel filters. SID 12-bit DAC quantization truncates to remove half-LSB DC bias.

Engine and Player Routing


Music Players (PTR/LEN/CTRL/STATUS/POSITION proto
Sound Engines
AY Player
0xF0C10
VGM/VGZ auto-convert
PSG / AY-3-8910
0xF0C00-0xF0C0F
3 Tone + Noise + Envelope
Ch 0-2
SN76489
0xF0C30-0xF0C3F
3 Tone + Noise
Ch 0-3
SID Player
0xF0E20
SID 6581/8580 x3
0xF0E00/0xF0E30/0xF0E50
3 Voices x 3 Chips = 9
Voices
SID1:Ch0-2 SID2:Ch4-6
SID3:Ch7-9
SAP Player
0xF0D10
POKEY
0xF0D00-0xF0D0A
4 Channels + Poly Counters
Ch 0-3
TED Player
0xF0F10
TED Audio
0xF0F00-0xF0F05
2 Voices (square + noise)
Ch 0-1
SoundChip
10 Channels
AHX Player
0xF0B84
AHX Replayer
0xF0B80-0xF0B91
4 Channels
Ch 0-3
Paula-style DMA shim (0xF2260-0xF22AF)
4 DMA Audio Channels
AROS audio.device
DAC bypass
MIDI/MUS Player
0xF0BA0
RawlandMini
MOD Player
0xF0BC0
+ FILTER_MODEL
WAV Player
0xF0BD8
+ POSITION

Audio Engine Plus Enhanced Mode

All five classic-style sound engines have a "Plus" enhanced mode, activated by writing `1` to their respective `PLUS_CTRL` register:

Engine	PLUS_CTRL Address	Enhancements
PSG+	`0xF0C20`	Enhanced render path: oversampling, filtering, drive/room shaping, stereo voicing
SID+	`0xF0E19`	Enhanced render path for SID voices (oversampling/filter/drive/room shaping)
POKEY+	`0xF0D09`	Enhanced render path (oversampling/filter/drive/room shaping)
TED+	`0xF0F05`	Enhanced render path plus TED-specific response shaping
AHX+	`0xF0B80`	AHX voice-state mapping with stereo spread/panning and room processing

The PSG uses the AY/YM logarithmic 16-step volume curve by default. A legacy linear curve is retained for older material that depends on it.

When Plus mode is enabled, the engine retains full backward compatibility with the standard register set while exposing additional capabilities. AHX maps tracker state to native SoundChip channels instead of producing an AROS audio DMA stream; AHX+ uses a 64-sample crossfade when enabling/disabling to prevent audio glitches.

AROS Paula-Style DMA Shim ABI

The AROS audio block at `0xF2260-0xF22AF` is an Intuition Engine shim for AROS `audio.device`, not a full Amiga Paula implementation.

- Guest code writes the shim with 32-bit aligned writes (`MOVE.L`). Narrower writes are outside the current ABI.
- A `DMACON` channel transition from `0` to `1` latches that channel's pointer, length, period, and volume. Active playback uses the latched values until the buffer ends.
- If the latched period or length is zero at arm time, the channel is not activated. The channel `DMACON` bit is cleared, the status bit is set, and a level-3 interrupt is raised when the corresponding `INTENA` bit is set.
- Buffer exhaust sets the status bit, raises a level-3 interrupt when the corresponding `INTENA` bit is set, marks the channel inactive, and clears that channel's `DMACON` bit. The guest must write `DMACON` enable again to start the next buffer.
- Out-of-range pointers beyond the active AROS profile RAM deactivate the channel, mute the DAC output, set the status bit, and raise a level-3 interrupt when the corresponding `INTENA` bit is set.
- Pointer writes ignore bit 0, length is a word count and preserves odd values, period writes with zero are ignored, and volume writes are clamped to `0..64`.

AROS HostFS DOS Handler ABI

The AROS DOS block at `0xF2220-0xF225F` is the MMIO command bridge used by the AROS m68k-ie packet handler. The guest writes up to four argument registers and then writes `AROS_DOS_CMD`; the Go-side `ArosDOSDevice` executes the request synchronously and returns AmigaDOS-style `RESULT1` and `RESULT2` values. `ADOS_CMD_EXAMINE_ALL` accelerates `ACTION_EXAMINE_ALL` through a 20-byte big-endian request descriptor, guest span validation, direct ExAllData packing, eac_LastKey continuation, and ERROR_ACTION_NOT_KNOWN fallback for match strings or hooks.

AROS HostFS fast paths use strict bulk guest-memory helpers, a 64 KiB sequential read-ahead cache, and cache invalidation on non-sequential reads, seeks outside cache, writes, truncates, close, create, delete, rename, and dirty close paths. Spans outside active guest RAM, high-pointer requests unsupported by the active bus, and low/high backing seam crossings fail closed. External host changes are best-effort and are not continuously watched.

AROS Host Socket ABI

The AROS host socket block at `0xF2500-0xF257F` backs the m68k-ie ROM `bsdsocket.library`. `0xF2400-0xF24FF` remains the SYSINFO ABI, so socket registers occupy the next 128-byte MMIO block. The AROS host socket block at 0xF2500-0xF257F uses 96-byte big-endian request descriptors, strict guest-memory helper copies, 64 KiB send/recv caps, 128-byte sockaddr caps, guest-visible descriptor handles, WaitSelect fd_set translation, Release/ReleaseCopy/Obtain transfer keys, and ENOSYS fail-closed behaviour when disabled. The guest writes `REQ_PTR` and `REQ_LEN`, places the descriptor in guest RAM, and triggers synchronous dispatch by writing `CMD`. Payload, fd_set, timeval, hostname, hostent, and sockaddr buffers are copied through those validated guest spans.

Supported command values are `1=socket`, `2=bind`, `3=listen`, `4=accept`, `5=connect`, `6=sendto`, `7=recvfrom`, `8=shutdown`, `9=setsockopt`, `10=getsockopt`, `11=getsockname`, `12=getpeername`, `13=ioctl`, `14=close`, `15=WaitSelect`, `16=gethostbyname`, `17=gethostbyaddr`, `18=gethostname`, `19=Dup2`, `20=GetEvents`, `21=ReleaseSocket`, `22=ReleaseCopyOfSocket`, and `23=ObtainSocket`. The Unix backend supports IPv4 TCP and UDP sockets, makes host file descriptors non-blocking, hides them behind guest-visible handles, maps `WaitSelect` fd_sets through that handle table, and rejects arbitrary emulator process descriptors with `EBADF`. `ReleaseSocket` removes the caller's guest descriptor and stores the host descriptor under a release key. `ReleaseCopyOfSocket` stores a duplicated host descriptor and leaves the caller's descriptor active. Passing release key `-1` asks the backend to allocate a positive key; `ObtainSocket` consumes that key and returns a new guest descriptor, or fails with `EWOULDBLOCK` when the key is absent. Raw sockets, packet filters, route manipulation, interface configuration, and monitoring APIs return unsupported errors. When networking is disabled or no backend is present, the block fails closed with `ENOSYS`.

Subsong Selection

SID, SAP, and AHX players support subsong selection for multi-tune files. Each player has a subsong register that selects which tune to play from a multi-song file.

SID PSID playback captures CIA1 timer-A latch writes at `$DC04/$DC05`; when non-zero, the player uses that latch as `cyclesPerTick`, so multispeed tunes run at `clockHz / latch`. SID MMIO opts into wide-write fanout: 16-bit and 32-bit writes are decomposed into little-endian byte register writes.

Generic Live-MIDI Port (Shared Subsystem)

The live-MIDI port is a CPU-agnostic MMIO device that feeds a raw running-status MIDI byte stream straight into the synth, distinct from the file player (`SOUND PLAY "x.mid"`) but sharing the same `MIDIEngine`, the same pool of 10 MIDI voices, and the same sound-chip mix key. There is one synth and one runtime-status owner; the live port and the file player never double-mix.

Three byte-wide registers drive it: `IE_MIDI_LIVE_DATA` (`0xF0BF4`, write a raw MIDI byte), `IE_MIDI_LIVE_STATUS` (`0xF0BF5`, read bit 0 = live port active), and `IE_MIDI_LIVE_CTRL` (`0xF0BF6`, write bit 0 = reset / all notes off). `midi_live.go` is a running-status state machine that turns the byte stream into `MIDIEvents` (note-on/off, control change, program change, pitch bend, including messages split across multiple writes) and applies them to the shared engine. Because the device hangs off the MachineBus, every core and EhBASIC reach it directly; the EhBASIC `MIDI NOTE`/`PROG`/`CTRL`/`SEND`/`RESET` keywords emit onto these registers. Live notes take precedence over the file player at voice allocation while active (a live note steals a file voice before another live voice) and clear on a CTRL reset. Per-architecture symbol names are declared in each SDK include (`ie32/ie64/ie65/ie68/ie80/ie86.inc`), windowed to each core's address convention. The Ebiten runtime status bar's `MIDI` legend lights when either the file player is playing or the live port is active (`MIDIEngine.LiveActive()`), so both MIDI sources share one indicator.

EmuTOS Atari MIDI ACIA bridge

EmuTOS does not know IE's native live-MIDI register map; it talks MIDI only through the Atari ST MIDI port, an **MC6850 ACIA**. `atari_midi_acia.go` is a minimal, output-only MC6850 shim (`AtariMIDIACIA`) that bridges that ACIA to the same `LiveMIDI`/`MIDIEngine` path, so notes from ST software or GEMDOS `Bconout(3, ...)` reach IE's synth. It is wired **only in EmuTOS mode** (`main.go`, guarded by `modeEmuTOS`) and holds no synth or voice state of its own: control writes forward to the LiveMIDI parser, status reads always report `TDRE` (ready to transmit), and there is no MIDI-in. The shim maps two address forms of the Atari contract `$FFFC04` (RS=0: control/status) and `$FFFC06` (RS=1: TX data / RX): the bus-canonical low-16 alias `0x0000FC04/06` (which the sign-extended guest access `0xFFFFFC04/06` is normalized to before handler lookup) and the 24-bit Atari hardware alias `0x00FFFC04/06` (which does not pass through sign-extension normalization and must be mapped explicitly). Both aliases use `MapIONoShadow` because they fall inside guest RAM and status/data accesses must not mirror handler values into guest memory. A control write of `ACIA_CTRL_MASTER_RESET` (CR1:CR0 == 11) calls `LiveMIDI.Reset()`. The IKBD ACIA at `$FFFC00/02` is deliberately **not** mapped: keyboard already flows through the IOREC pump in `emutos_loader.go`. This bridge is EmuTOS-specific; there is no generic ACIA emulation for other modes.

6. Memory Map

All CPU cores observe the same guest physical address space. Address ranges in this table are not per-core private maps unless explicitly stated. Some ranges have multiple architectural uses: for example, a decoded RAM range may also contain conventional worker buffers, framebuffer memory, or OS/profile-owned allocations. In those cases, the table describes reservations within the shared physical map, not separate mappings.

The `Decoded as` column describes how the bus routes a physical access. The `Architectural use` and `Owner / lifetime` columns describe the guest contract or current subsystem reservation inside that decoded mapping. Overlapping rows are intentional when a reservation lives inside a broader shared-RAM range.

Range	Size	Decoded as	Architectural use	Visible to	Owner / lifetime	Notes
`0x00000-0x9EFFF`	636KB	Shared RAM	Main low-memory programme/data area	All CPU cores	Guest and boot-profile convention	Includes IE32 vector table base at `0x00000` and programme load convention at `0x01000`.
`0x9F000-0x9FFFF`	4KB	Shared RAM	Default stack seed region	All CPU cores	CPU reset convention	IE32 and IE64 seed stack pointers at `0x9F000`; it is still ordinary shared RAM.
`0xA0000-0xAFFFF`	64KB	VGA VRAM window	VGA graphics aperture	All CPU cores	VGA subsystem	x86 can also access this as conventional VGA graphics memory.
`0xB8000-0xBFFFF`	32KB	VGA text buffer	VGA text aperture	All CPU cores	VGA subsystem	Conventional text-memory range.
`0xD0000-0xDFFFF`	64KB	Voodoo texture memory	Voodoo texture upload/storage window	All CPU cores	Voodoo subsystem	Shared physical texture-memory aperture.
`0xF0000-0xF049B`	1180B	MMIO	VideoChip, copper, blitter, palette, and extended blitter registers	All CPU cores	Video subsystem	Layer-0 video control and DMA-style graphics operations.

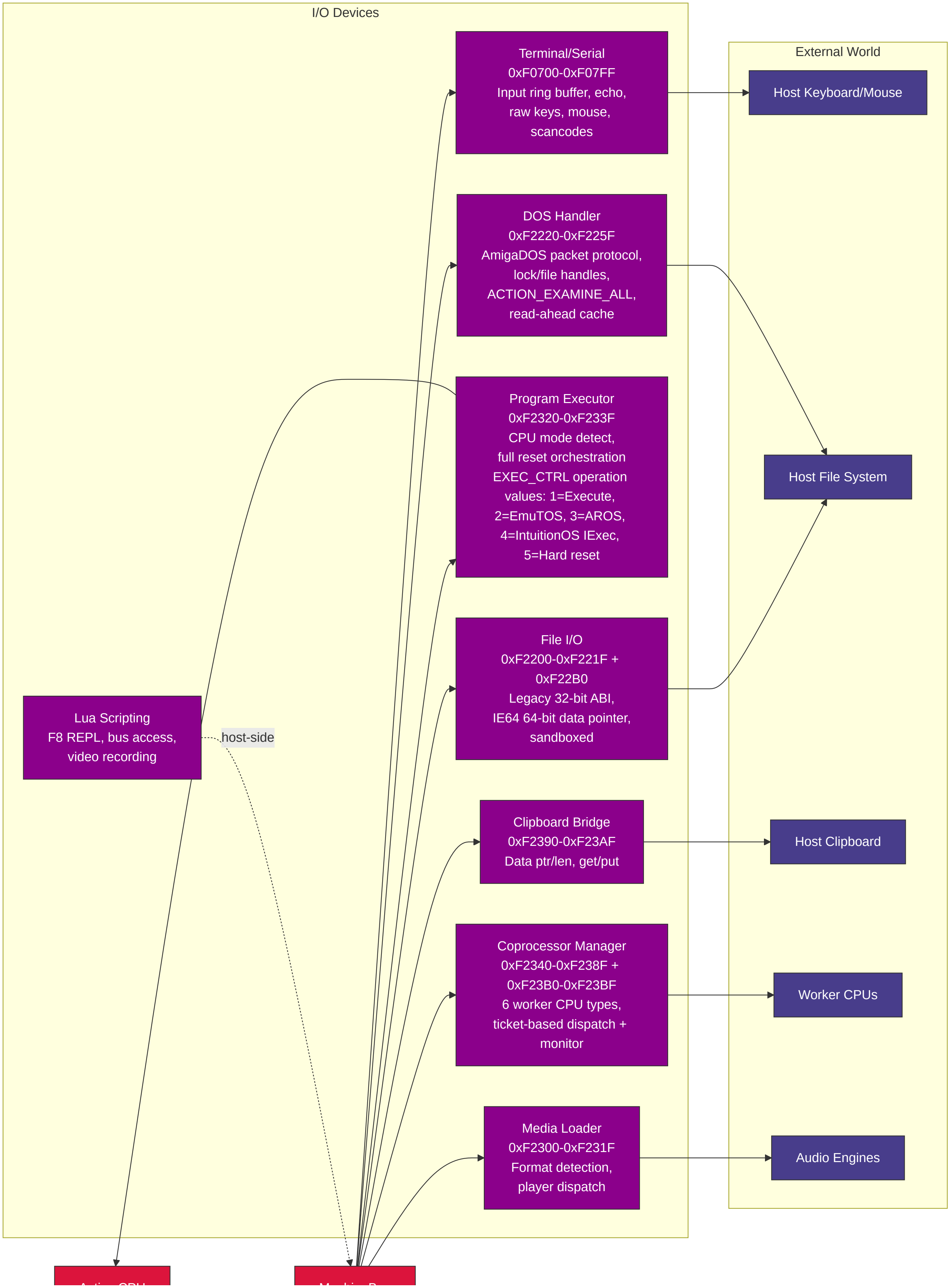
Range	Size	Decoded as	Architectural use	Visible to	Owner / lifetime	Notes
`0xF0700-0xF07FF`	256B	MMIO	Terminal, serial, mouse, keyboard, and `RTC_EPOCH` (`0xF0750`)	All CPU cores	I/O subsystem	Input, terminal, and clock-facing low MMIO.
`0xF0800-0xF0B7F`	896B	MMIO	SoundChip channels, including FLEX	All CPU cores	Audio subsystem	Runtime audio-chip register block.
`0xF0B80-0xF0B91`	18B	MMIO	AHX engine/player	All CPU cores	Audio subsystem	Player-facing register block.
`0xF0BA0-0xF0BBF`	32B	MMIO	MIDI/MUS player	All CPU cores	Audio subsystem	Player-facing register block.
`0xF0BC0-0xF0BD7`	24B	MMIO	MOD player	All CPU cores	Audio subsystem	Player-facing register block.
`0xF0BD8-0xF0BF3`	28B	MMIO	WAV player	All CPU cores	Audio subsystem	Player-facing register block.
`0xF0BF4-0xF0BF6`	3B	MMIO	Generic live-MIDI port	All CPU cores	Audio subsystem	Byte-wide running-status MIDI stream: data write, active-status read, and reset control.
`0xF0C00-0xF0C0F`	16B	MMIO	PSG engine (AY-3-8910/YM2149 registers)	All CPU cores	Audio subsystem	Register-select/data-compatible PSG block.
`0xF0C10-0xF0C1F`	16B	MMIO	PSG / AY player	All CPU cores	Audio subsystem	Player-facing register block.
`0xF0C20`	1B	MMIO	PSG+ control	All CPU cores	Audio subsystem	Extended PSG control byte.
`0xF0C30-0xF0C3F`	16B	MMIO	Native SN76489 latch/data, ready, and LFSR mode registers	All CPU cores	Audio subsystem	Native SN76489-compatible control block.
`0xF0C40-0xF0CFF`	192B	MMIO	SID2 flex-style SoundChip channels 4-6	All CPU cores	Audio subsystem	Multi-SID/FLEX channel range.
`0xF0D00-0xF0D0A`	11B	MMIO	POKEY engine	All CPU cores	Audio subsystem	POKEY register block.
`0xF0D10-0xF0D20`	17B	MMIO	SAP player	All CPU cores	Audio subsystem	Player-facing register block.
`0xF0D40-0xF0DFF`	192B	MMIO	SID3 flex-style SoundChip channels 7-9	All CPU cores	Audio subsystem	Multi-SID/FLEX channel range.
`0xF0E00-0xF0E1C`	29B	MMIO	SID1 engine (6581/8580)	All CPU cores	Audio subsystem	Primary SID register block.
`0xF0E20-0xF0E2D`	14B	MMIO	SID player	All CPU cores	Audio subsystem	Player-facing register block.

Range	Size	Decoded as	Architectural use	Visible to	Owner / lifetime	Notes
`0xF0E30-0xF0E4C`	29B	MMIO	SID2 engine (6581/8580)	All CPU cores	Audio subsystem	Secondary SID register block.
`0xF0E50-0xF0E6C`	29B	MMIO	SID3 engine (6581/8580)	All CPU cores	Audio subsystem	Tertiary SID register block.
`0xF0E80-0xF0EFF`	128B	MMIO	SoundChip SFX trigger channels	All CPU cores	Audio subsystem	Trigger-oriented SFX range.
`0xF0F00-0xF0F05`	6B	MMIO	TED audio engine	All CPU cores	Audio subsystem	TED audio registers.
`0xF0F10-0xF0F1F`	16B	MMIO	TED player	All CPU cores	Audio subsystem	Player-facing register block.
`0xF0F20-0xF0F6B`	76B	MMIO	TED video	All CPU cores	TED video subsystem	Stride-4 TED video register mapping.
`0xF3000-0xF6FFF`	16KB	MMIO / TED VRAM aperture	TED private video RAM	All CPU cores	TED video subsystem	Bus-routed TED character, colour, and screen memory.
`0xF1000-0xF13FF`	1KB	MMIO	VGA registers	All CPU cores	VGA subsystem	Sequencer, CRTC, graphics-controller, attribute-controller, and DAC state.
`0xF1400-0xF140F`	16B	MMIO	Host helper	All CPU cores	Host-helper subsystem	Security-sensitive command helper gated by runtime configuration.
`0xF2000-0xF2017`	24B	MMIO	ULA registers	All CPU cores	ULA subsystem	Includes paged VRAM data port.
`0xFA000-0xFBAFF`	6912B	ULA VRAM aperture	ULA display memory	All CPU cores	ULA subsystem	Separate decoded aperture for ULA VRAM access.
`0xF2100-0xF213F`	64B	MMIO	ANTIC registers	All CPU cores	ANTIC subsystem	ANTIC display-list and DMA control block.
`0xF2140-0xF21FB`	188B	MMIO	GTIA registers	All CPU cores	GTIA subsystem	GTIA colour, player/missile, priority, and collision state.
`0xF2200-0xF221F`	32B	MMIO	File I/O	All CPU cores	File-I/O bridge	Legacy host-file bridge register block with `32`-bit name, data, length, status, error, and read-cap registers.
`0xF22B0-0xF22B7`	8B	MMIO64	File I/O IE64 extension	IE64	File-I/O bridge	`FILE_DATA_PTR64`, a `64`-bit data-buffer pointer for IE64 code that reads or writes host files from high guest RAM. The legacy File I/O block remains unchanged.
`0xF2220-0xF225F`	64B	MMIO	AROS DOS handler	All CPU cores	AROS profile	AROS DOS integration register block.
`0xF2260-0xF22AF`	80B	MMIO	AROS Paula-style DMA shim	All CPU cores	AROS profile	Audio/DMA compatibility shim.
`0xF2300-0xF231F`	32B	MMIO	Media loader	All CPU cores	Media-loader subsystem	Format-agnostic media loading control block.

Range	Size	Decoded as	Architectural use	Visible to	Owner / lifetime	Notes
`0xF2320-0xF233F`	32B	MMIO	Program executor	All CPU cores	Program executor	CPU/program switching control block.
`0xF2340-0xF238F`	80B	MMIO	Coprocessor manager	All CPU cores	Coprocessor subsystem	Worker lifecycle, dispatch, and ticket registers.
`0xF2390-0xF23AF`	32B	MMIO	Clipboard bridge	All CPU cores	Clipboard bridge	Host clipboard integration register block.
`0xF23B0-0xF23BF`	16B	MMIO	Coprocessor extended monitor registers	All CPU cores	Coprocessor subsystem	Per-worker monitor/status extension.
`0xF23C0-0xF23DF`	32B	MMIO	AROS IRQ diagnostic registers	AROS M68K profile while AROS loader is active	AROS profile	Read-only diagnostic MMIO mapped by the AROS loader and unmapped during AROS DMA teardown; not a universal interrupt-controller ABI.
`0xF23E0-0xF23FF`	32B	MMIO	Bootstrap HostFS	All CPU cores	Bootstrap profile	HostFS boot helper register block.
`0xF2400-0xF24FF`	256B	MMIO	SYSINFO RAM-size discovery	All CPU cores	System information ABI	Reports total and active visible RAM.
`0xF2500-0xF257F`	128B	MMIO	AROS host socket bridge	AROS M68K profile	AROS profile	Host-backed `bsdsocket.library` command bridge.
`0xF8000-0xF87FF`	2KB	MMIO	Voodoo 3D registers and palette	All CPU cores	Voodoo subsystem	3D control, state, and palette register block.
`0xF8140-0xF823F`	256B	MMIO	Voodoo fog table	All CPU cores	Voodoo subsystem	64 entries x 4 bytes.
`0x100000-0x5FFFFF`	5MB	Shared RAM / VRAM-backed region	Main video framebuffer and graphics-visible memory	All CPU cores	Video subsystem plus guest convention	Subranges may be reserved for coprocessor worker buffers.
`0x200000-0x27FFFF`	512KB	Shared RAM	IE32 worker area	All CPU cores	Coprocessor convention	Lies inside the graphics-visible shared-memory range; not a separate address space.
`0x280000-0x2FFFFF`	512KB	Shared RAM	M68K worker area	All CPU cores	Coprocessor convention	Lies inside the graphics-visible shared-memory range; not a separate address space.
`0x300000-0x30FFFF`	64KB	Shared RAM	6502 worker area	All CPU cores	Coprocessor convention	Lies inside the graphics-visible shared-memory range; not a separate address space.
`0x310000-0x31FFFF`	64KB	Shared RAM	Z80 worker area	All CPU cores	Coprocessor convention	Lies inside the graphics-visible shared-memory range; not a separate address space.

Range	Size	Decoded as	Architectural use	Visible to	Owner / lifetime	Notes
`0x320000-0x39FFFF`	512KB	Shared RAM	x86 worker area	All CPU cores	Coprocessor convention	Lies inside the graphics-visible shared-memory range; not a separate address space.
`0x3A0000-0x41FFFF`	512KB	Shared RAM	IE64 worker area	All CPU cores	Coprocessor convention	Lies inside the graphics-visible shared-memory range; not a separate address space.
`0x790000-0x7917FF`	6KB	Shared RAM	Coprocessor mailbox ring buffers	All CPU cores	Coprocessor subsystem	Shared request/completion rings; outside the main graphics-visible range.
`0x800000-0x80FFFF`	64KB	Shared RAM	Media-loader staging buffer	All CPU cores	Media-loader subsystem	Reservation at the base of the AROS fast-memory range when that profile is active.
`0x800000-0x1DFFFFF`	22MB	Shared RAM	AROS fast memory	All CPU cores	AROS profile	Profile-owned allocation range, not a distinct per-core map.
`0x1E00000-0x5DFFFFF`	64MB	Shared RAM / video-profile memory	AROS video RAM	All CPU cores	AROS profile / video subsystem	Profile-owned video allocation range within shared physical memory.

7. I/O Peripherals


I/O Devices
External World
Terminal/Serial
0xF0700-0xF07FF
Input ring buffer, echo,
raw keys, mouse,
scancodes
Host Keyboard/Mouse
DOS Handler
0xF2220-0xF225F
AmigaDOS packet protocol,
lock/file handles,
ACTION_EXAMINE_ALL,
read-ahead cache
Program Executor
0xF2320-0xF233F
CPU mode detect,
full reset orchestration
EXEC_CTRL operation
values: 1=Execute,
2=EmuTOS, 3=AROS,
4=IntuitionOS IExec,
5=Hard reset
Host File System
File I/O
0xF2200-0xF221F +
0xF22B0
Legacy 32-bit ABI,
IE64 64-bit data pointer,
sandboxed
Lua Scripting
F8 REPL, bus access,
video recording
host-side
Clipboard Bridge
0xF2390-0xF23AF
Data ptr/len, get/put
Host Clipboard
Coprocessor Manager
0xF2340-0xF238F +
0xF23B0-0xF23BF
6 worker CPU types,
ticket-based dispatch +
monitor
Worker CPUs
Media Loader
0xF2300-0xF231F
Format detection,
player dispatch
Audio Engines

The File I/O ABI keeps the original 32-bit register block at `0xF2200-0xF221F` for every CPU. IE64 adds `FILE_DATA_PTR64` at `0xF22B0` as a separate 64-bit MMIO register for data buffers in high guest RAM, including buffers allocated from the native AOT arena. Non-IE64 software does not need to know about the extension.

Coprocessor Worker Dispatch

The coprocessor manager supports 6 worker CPU types (IE32, IE64, 6502, M68K, Z80, x86) with ticket-based job dispatch and mailbox ring buffers at `0x790000`. Each worker type has a dedicated shared-memory reservation in the unified physical map (see memory map above), not a private per-core address space. The main CPU enqueues work via MMIO writes; workers execute independently and post results back through their mailbox slots. Each ring has 16 descriptor slots but uses one slot to distinguish full from empty, so it can hold 15 queued requests at once. When `COPROC_IRQ_CTRL` bit 0 is set and an M68K IRQ target plus completion watcher have been configured, the coprocessor asserts M68K interrupt level 6 on job completion, with the finished ticket ID readable from `COPROC_COMPLETED_TICKET`. Non-M68K modes should observe completion through `POLL` or `WAIT`.

Lua Scripting

The Lua scripting engine (`script_engine.go`) runs in its own goroutine and provides host-side automation over the shared 32-bit bus/MMIO surface plus CPU-adapter debugger APIs. Its `mem.*` helpers are raw 32-bit bus helpers, not an above-4GiB IE64 RAM or CPU-virtual-address API. It supports an F8 REPL for interactive debugging, video recording, and direct chip register manipulation. Scripts use the `.ies` extension and are loaded via the IE Script Engine.

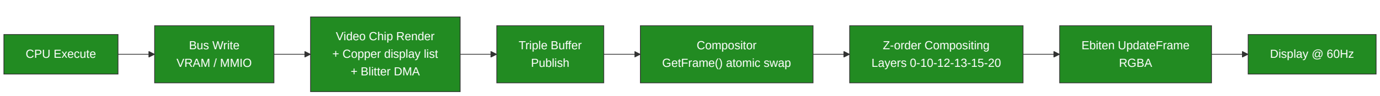
IEMon Reverse-Debug Snapshot Contract

IEMon whole-machine snapshots enumerate the monitor CPU registry by stable CPU id and label, so profiles with omitted CPUs or multiple coprocessors restore by identity rather than by CPU type. Shared bus RAM and IE64 backing memory are stored as sparse 4 KiB pages. The reverse-history chain keeps full sparse checkpoints plus page deltas anchored to retained checkpoints; the monitor materialises deltas before `rg` or `rt` applies a restore.

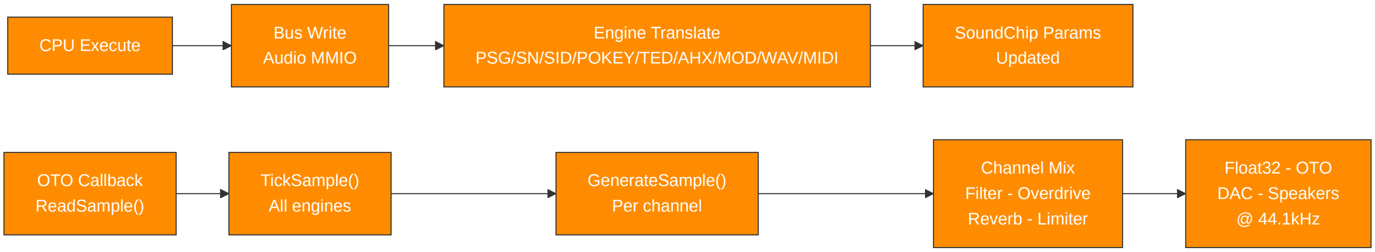
Mutable devices join the snapshot contract through `MachineMonitor.RegisterSnapshotDevice`. Each device supplies a stable name, a version, and an opaque guest-visible state blob. Restore fails closed if a captured device is absent, preventing partial reverse-debug restores. The production monitor registers the main video chip, sound chip, terminal MMIO, command-style host helpers, compatibility audio/video engines, and AROS clipboard/audio-DMA bridges when present; new guest-visible timers, DMA engines, IRQ controllers, and host bridges must register their own versioned blobs before they are considered covered by whole-machine reverse debugging.

8. Data Flow

Video Pipeline



Audio Pipeline



CPU Mode Switching

9. Concurrency Model and System Timing

Goroutine Inventory

Goroutine	Clock Source	Rate	Synchronisation
CPU execution	Free-running `for running.Load()` loop	As fast as Go scheduler allows	MMIO writes invoke I/O callbacks under per-chip `mu`
VideoChip refresh	`time.NewTicker`	60Hz	Copper + blitter + dirty-tile copy + buffer swap
VGA render	`time.NewTicker`	60Hz	Renders into triple-buffer slot, publishes via `sharedIdx.Swap()`
ULA render	`time.NewTicker`	60Hz	Same triple-buffer pattern
TED render	`time.NewTicker`	60Hz	Same triple-buffer pattern
ANTIC render	`time.NewTicker`	60Hz	Same triple-buffer pattern
Compositor	`time.NewTicker`	60Hz	Calls `GetFrame()` (atomic swap) on each source, blends, sends to display
OTO audio thread	Host audio hardware callback	44.1kHz	Calls `SoundChip.ReadSample()` -> `GenerateSample()` directly
Terminal output	`time.NewTicker`	100Hz	Flushes output buffer to host terminal
Coprocessor workers	On-demand	Varies	Independent CPUs with mailbox ring buffers
Script engine	Event-driven	Varies	Lua goroutine with frame-sync channel

Three Independent Clocks

```
CPU:   Free-running (no fixed clock -- instruction throughput varies with host speed)
Video: 6 video sources plus compositor tick at 60Hz; non-VideoChip sources use lock-free triple-buffer handoff
Audio: OTO hardware callback drives sample generation at 44.1kHz -- no IE-owned goroutine
```

Synchronisation Model

Hot paths (lock-free): - `atomic.Bool` - CPU `running`, chip `enabled`, `compositorManaged` - `atomic.Int32` - triple-buffer `sharedIdx` - `atomic.Pointer` - selected CPU pointer

Cold paths (mutex-protected): - `sync.Mutex` (`chip.mu`, `video.mu`) - configuration changes, setup, stop

CPU-to-Video: - CPU writes to VRAM/MMIO invoke I/O callbacks under per-chip `mu` - Video chips read VRAM lock-free (snapshot-render pattern: copy under lock, render without lock)

CPU-to-Audio: - CPU writes to audio MMIO update channel parameters under `chip.mu` - OTO callback reads parameters atomically or under `chip.mu` for `GenerateSample()`

No CPU-video vsync coupling: - CPU runs ahead freely - no wait-for-vblank blocking - WAIT register polls `vblankActive atomic.Bool` without blocking the CPU

Appendix: Key Source Files

File	Role
`registers.go`	Master I/O address map - all region boundaries
`machine_bus.go`	MachineBus: autodetected guest RAM, MapIO, ioPageBitmap, Read/Write, SYSINFO accessors
`memory_sizing.go`	Boot-time guest RAM autodetection: total guest RAM + active visible RAM with platform reserves
`profile_bounds.go`	Source-owned profile bounds for EmuTOS, AROS, IE64 BASIC
`emulator_cpu.go`	EmulatorCPU interface definition
`video_interface.go`	VideoSource, VideoOutput, ScanlineAware interfaces
`video_compositor.go`	Compositor pipeline, Z-order blending
`video_chip.go`	VideoChip + Copper + Blitter + Palette
`video_vga.go`	VGA engine (Sequencer, CRTC, GC, AC, DAC)
`video_ula.go`	ULA engine (Spectrum display)
`video_ted.go`	TED video engine
`video_antic.go`	ANTIC + GTIA engines
`video_voodoo.go`	Voodoo 3D pipeline
`audio_chip.go`	SoundChip (10-channel synthesis)
`psg_engine.go`	PSG / AY-3-8910
`sn76489_chip.go`	SN76489 programmable sound generator
`sid_engine.go`	SID 6581/8580
`pokey_engine.go`	POKEY
`ted_engine.go`	TED audio
`ahx_engine.go`	AHX replayer
`mod_player.go`	MOD player

File	Role
`wav_player.go`	WAV player
`midi_player.go` / `midi_engine.go` / `midi_parser.go`	SMF `.mid`/`.midi` and Doom `.mus`; fixed RawlandMini IE SoundChip GM-style/chiptune interpretation, 10 active MIDI voices with deterministic stealing
`midi_live.go`	Generic live-MIDI MMIO port: running-status byte-stream parser feeding the shared MIDIEngine (all cores + BASIC)
`atari_midi_acia.go`	EmuTOS-only MC6850 MIDI ACIA shim ($FFFC04/06, low-16 + 24-bit aliases) bridging output-only MIDI bytes into the shared LiveMIDI/MIDIEngine
`cpu_ie32.go`	IE32 CPU
`cpu_ie64.go`	IE64 CPU + interpreter
`fpu_ie64.go`	IE64 FPU (16 x float32)
`jit_emit_arm64.go`	IE64 JIT ARM64 emitter
`jit_emit_amd64.go`	IE64 JIT x86-64 emitter
`cpu_m68k.go`	M68K 68020
`fpu_m68881.go`	M68881 FPU (8 x float64)
`cpu_z80.go`	Z80
`cpu_z80_runner.go`	Z80 bus adapter, port I/O bridge, bank windows
`cpu_six5go2.go`	6502 + bus adapter, ioTable dispatch, bank windows
`cpu_x86.go`	x86
`cpu_x86_runner.go`	x86 bus adapter, shared port I/O bridge, bank windows, and VGA VRAM window
`fpu_x87.go`	x87 FPU (8 x float64 stack)
`terminal_io.go`	Terminal / Serial / Mouse / Keyboard
`file_io.go`	File I/O
`program_executor.go`	Program Executor
`coprocessor_manager.go`	Coprocessor Manager
`clipboard_bridge.go`	Clipboard Bridge
`media_loader.go`	Media format detection + player dispatch
`script_engine.go`	Lua scripting engine
`aros_loader.go`	AROS ROM boot manager
`aros_dos_intercept.go`	AmigaDOS packet handler (MMIO)
`aros_audio_dma.go`	AROS Paula-style DMA shim